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The Narrative Flow of the Project

The project follows a problem-answer-architecture narrative, directly related to how the Scrum technique is used. The problem statement is divided into sub-questions, each with its own smaller questions and tasks, resulting in a three-layer structure for the project.

The project problem statement is the main layer that is not explicitly highlighted in each part of the structure but serves as the overall theme. The second layer is the major question that needs to be answered, which drives the project forward and acts as a key component on its own.

The major question is further divided into four or five sub-questions, and each sub-question involves a small task that is saved as a view, or what I call checkpoints. In the Scrum technique, I do not build a large, robust structure in one go; instead, I work on smaller tasks that help improve understanding and identify key problems in specific areas of the project.

So, the project progresses through tasks, and these tasks are mainly saved as views, which act as checkpoints to validate any changes, whether new data is added in the future or refinements are needed in any of the sub-questions. These views allow me to work on a specific sub-question without damaging or significantly affecting the overall structure.

In the end, the project is structured in a way that keeps the analysis modular and flexible, where each step can be validated independently while still contributing to the larger problem, making the entire system easier to manage, test, and refine.

Problem Statement of the Project

I conducted a machine learning project where I focused on index returns as the target, using five stocks that are the major constituents of that index as features. The aim was to understand the underlying themes and patterns. The main takeaway was that increasing the complexity of the model did not improve the understanding of the dataset or increase its usefulness. From this, I realized that the problem was not with the model itself, but with the limitations of the features I used.

In reality, stock returns are part of a cause-and-effect chain, where the first layer is how market participants perceive the stock. This unstructured data includes news related to the stock, broader macroeconomic factors, and earnings-related information, all of which influence stock returns.

The project essentially asks whether adding these unstructured data as features can provide meaningful additional insights and improve the understanding of the dataset, rather than focusing only on model complexity. In other words, can incorporating unstructured data such as news, macroeconomic information, and earnings complement stock return data to better understand market behaviour, and if they can, what are the limitations and challenges of using such data?

One major challenge faced during this process is related to data availability. Since the dataset spans from November 2015 to October 2025, it is difficult to obtain complete historical news and earnings data. Therefore, the dataset is constructed by focusing on the most relevant 20–30 news items for each stock and major global economic events, along with 1–2 earnings summaries per year for each company. The stocks that I am focusing on are TCS, Reliance, Infosys, ICICI, and HDFC, along with the index Nifty 50.

Overall, the project shifts the focus from improving model complexity to improving feature quality, by exploring whether unstructured data can add meaningful context to stock returns, while also understanding the limitations that come with such data.

First Major Question of the Project -

I have a dataset that includes news and stock earnings. In real life, there are short-term and long-term aspects. Short-term typically focuses on daily updates, while long-term looks at progress over a longer period. This can be seen with news and earnings: news comes daily and reflects real-time information about the stock or the global economy, whereas earnings reflect a longer pattern, summarizing past performance and future expectations of the company.

Both operate differently. Since the stock market functions daily, it is more influenced, from a human perspective, by the continuous flow of news that market participants consume regularly.

So, the first question is whether the sentiment or emotion derived from financial news or global economic news provides any meaningful or useful signal to understand market behaviour. In simple terms, do the sentiments generated by financial news offer valuable insights into market movement?

The main problem, or challenge, that I already mentioned is the sparsity of the news that I have for the last 10 years. While I can get daily market data, obtaining consistent historical news data is not possible. So, the question shifts toward major news days: whether the sentiment or emotion derived from financial news on those specific days provides a meaningful signal to understand market behaviour or not.

Overall, the question focuses on testing whether sentiment extracted from limited but important news events can act as a useful signal for market behaviour, given the constraints in data availability.

Sub Question 1 -

The first question before understanding the pattern of sentiments derived from financial news for market behaviour is how these news data or text can be converted into usable sentiment signals. If I am unable to convert them into usable sentiment signals, understanding market behaviour with them will be more challenging or could even fail. So, the first step is to build the foundation—how I can convert these news data into usable signals.

For this, the first thing I need to understand is what the meaning of sentiment is. Sentiment refers to the tone that a sentence conveys, the feeling someone perceives while reading it. Each sentence generally has three main tones: positive, neutral, and negative. This is what I aim to extract from the sentiment. But how am I going to do that?

For this, I will use NLP, which is the main process in this project. NLP, or Natural Language Processing, simply means helping computers understand and extract meaningful information from unstructured text data. I know this isn't in very simple terms, but basically it involves transforming text data into numerical values that can be measured based on different language patterns. One of the patterns I am currently using is sentiment.

In NLP, this creates two numerical outputs: one is the sentiment score, also known as the polarity score, which ranges from -1 to +1. This score indicates the direction of the sentiment in the text—closer to +1 for positive, closer to -1 for negative, and near 0 for neutral. The other output is the magnitude, or the strength, which shows the intensity of the tone. It ranges from 0 to 1 in this case. This strength metric indicates how strongly the tone is conveyed; for example, the news might be positive but not presented with a strongly affirmative tone. Using these two outputs, I can extract measurable and meaningful insights.

The main challenge arises when extracting polarity scores and magnitudes for sentiment analysis. Since I am using BigQuery SQL, which does not directly support the NLP functions needed for this process, I switch to Python and use the Vader sentiment analyzer. This Vader tool is a rule-based NLP model, meaning it uses predefined rules and a dictionary of words to assign values when data enters it.

Vader uses its own dictionary to analyze each sentence, breaking it down into words and providing four scores: positive, negative, neutral, and compound. The compound score reflects the overall emotion in the sentence, which I use as the polarity score for sentiment analysis because it offers a complete picture of the overall sentiment and direction. For magnitude, I combine the positive and negative scores, which helps to understand the intensity of the words in the sentence.

I extract these scores, upload the CSV file into BigQuery SQL, and then create a view for them, as the main analysis will be performed in BigQuery SQL.

In the end, this step focuses on building a structured foundation by converting unstructured text into measurable sentiment signals, ensuring that further analysis on market behaviour can be performed in a consistent and quantifiable way.

	Date	Source	news_data	Sentiment_score	Sentiment_magnitude
2	14-02-2015	Reuters	HDFC Bank reported quarterly net profit growth of around 20%, supported by strong retail and corporate loan demand, highlighting res	0.8689	0.431
3	21-04-2016	Business S	HDFC Bank announced strong quarterly earnings with stable margins and low NPAs, reinforcing its conservative lending model while t	0.5423	0.352
4	08-11-2016	Reuters	Following demonetisation announced by the Indian government, HDFC Bank witnessed a surge in deposits and digital transactions as	0	0
5	13-01-2017	Mint	HDFC Bank reported steady growth driven by expansion in rural and semi-urban markets, focusing on retail lending and maintaining st	0.7717	0.343
6	17-07-2017	Economic	HDFC Bank delivered consistent quarterly performance with strong retail loan growth and stable margins, continuing to outperform pe	0.7783	0.349
7	20-04-2018	Reuters	HDFC Bank posted strong earnings growth with stable asset quality and improving margins, driven by expansion in retail lending and di	0.8834	0.413
8	01-08-2018	Business S	HDFC Bank became Indiaâ€™s most valuable lender by market capitalization, reflecting strong investor confidence in its consistent pr	0.9286	0.4
9	14-01-2019	Economic	HDFC Bank reported strong quarterly earnings supported by robust retail loan growth and stable asset quality, maintaining leadership	0.9231	0.41
10	20-04-2019	Reuters	HDFC Bank maintained steady profit growth and stable margins despite macroeconomic challenges, supported by strong retail lendin	0.8864	0.483
11	21-10-2019	Mint	HDFC Bank continued expanding its digital banking services, focusing on improving customer experience and increasing transaction v	0.6597	0.197
12	27-03-2020	Reuters	HDFC Bank announced support measures during COVID-19 disruptions, including loan moratoriums and liquidity assistance, aiming t	0.128	0.324
13	18-07-2020	Economic	Despite pandemic-related disruptions, HDFC Bank reported resilient earnings and stable asset quality, demonstrating strong risk mar	0.7867	0.389
14	02-12-2020	Reuters	RBI barred HDFC Bank from issuing new credit cards after repeated digital outages, highlighting operational risks and forcing the bank	0.4215	0.212
15	15-01-2021	Business S	HDFC Bank initiated major investments in digital infrastructure and compliance improvements following RBI restrictions, aiming to res	0.8316	0.299
16	23-02-2021	Reuters	HDFC Bank invested in a payments entity promoted by Tata Group, marking strategic expansion into Indiaâ€™s digital payments ecosys	0.7184	0.188
17	27-05-2021	Reuters	RBI imposed a monetary penalty on HDFC Bank for irregularities in its auto loan portfolio, highlighting compliance gaps and reinforcing	0	0.219
18	04-04-2022	Reuters	HDFC Ltd announced a merger with HDFC Bank in Indiaâ€™s largest financial sector deal, aiming to combine housing finance and ban	0.6249	0.159
19	01-07-2023	Reuters	HDFC Bank completed its merger with HDFC Ltd, creating one of the worldâ€™s largest banks with expanded balance sheet, increase	0.7096	0.204
20	13-03-2023	Business S	A data breach at HDB Financial Services, a subsidiary of HDFC Bank, exposed sensitive information of millions of customers, raising c	-0.34	0.108
21	13-11-2023	Reuters	RBI retained HDFC Bank in its list of domestic systemically important banks, requiring higher capital buffers and reinforcing its â€œtoc	0.2263	0.097
22	10-04-2024	Mint	HDFC Bank expanded its physical presence by opening new branches in underserved regions, strengthening distribution network and	0.7184	0.188
23	30-10-2024	Reuters	HDB Financial Services, a subsidiary of HDFC Bank, filed for a $1.5 billion IPO to meet regulatory requirements, supporting capital exp	0.6908	0.174
24	15-09-2024	Reuters	RBI imposed a penalty on HDFC Bank for non-compliance with deposit rate regulations and customer service standards, highlighting o	-0.7186	0.219
25	19-04-2025	Republic B	HDFC Bank reported strong FY2025 results with rising net profit and net interest income, along with announcing a dividend, reflecting	0.8773	0.352
26	19-07-2025	Mint	HDFC Bank scheduled board meeting to review quarterly results and consider dividend and bonus issuance, signaling continued focus	0.5423	0.123
27	15-09-2025	Reuters	Dubai regulator restricted a branch of HDFC Bank from onboarding new clients due to compliance deficiencies, raising concerns abo	-0.3818	0.094

news_with_sentiment

Sub Question 2 -

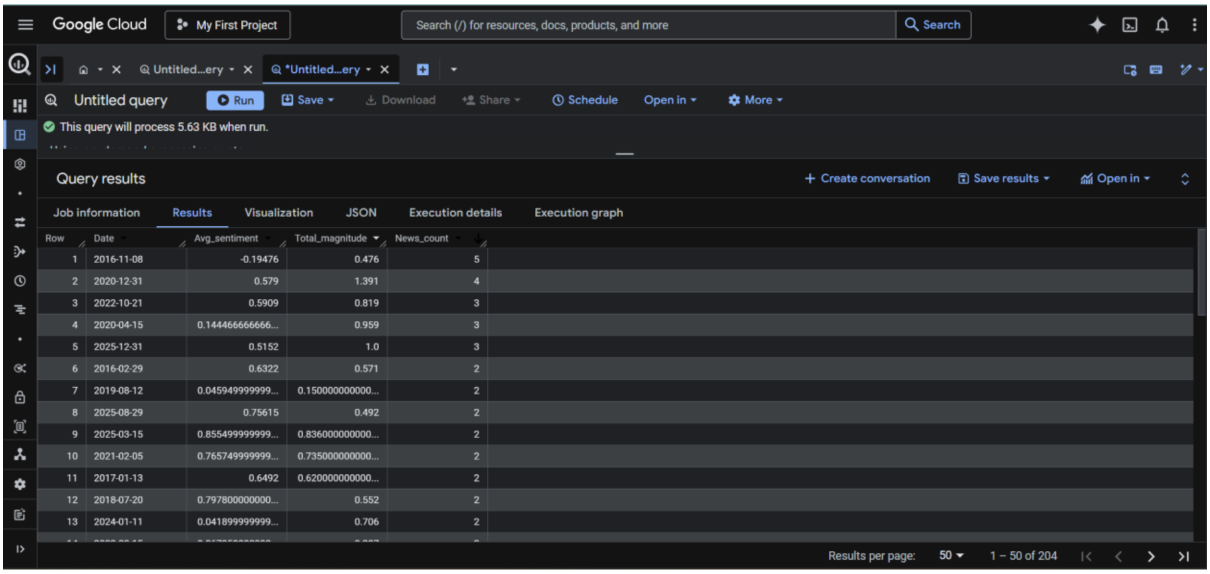
After calculating the polarity or sentiment score and magnitude score for each news item, there is something that needs to be done. Since some days may have multiple news articles, understanding the overall mood of that day becomes important. Additionally, uniformity matters because, in the market returns dataset, each date is represented by a single row. Therefore, when analyzing market behaviour based on these news days, maintaining uniformity is crucial. So, how can I implement or achieve this uniformity?

One of the methods is aggregation, which uses functions to combine multiple rows of data into a summarized entry or to transform multiple entries into one major entry. Summarizing data at the date level helps me understand the overall tone for that specific date. This aggregation converts multiple news-level observations into a single daily-level feature, making it compatible with the market dataset.

To aggregate this, in SQL, it is done by grouping data by Date. I plan to use the average function for the sentiment score, as it is important because it indicates the overall mood or tone of that day. For the magnitude score, I will use the sum, since magnitude acts as a relative measure compared to other days, helping to identify which days had higher intensity and which did not.

So, when I implemented the grouping by date along with these aggregation functions, there were many dates with multiple news items that needed to be combined. The main challenge, as I mentioned earlier, is the scarcity of news because the dataset spans from November 2015 to October 2025. This 10-year period makes it difficult to gather enough news, but this aggregation step will be helpful in the future when current data can be added, and recent news can be more easily incorporated.

Overall, this step ensures uniformity by converting multiple news entries into a single daily-level signal, making the sentiment data compatible with market returns while also preparing the dataset for future scalability.



Sub Question 3 -

Now, after grouping the data and applying aggregation functions to the sentiment score and the magnitude score, the main question arises once uniformity with respect to the market return data is achieved. Can we observe any alignment between the news data and the market return data? Is the tone of the news data aligned with the market return value or not? And how am I going to do it?

So, first of all, I need to focus on two terms. The first is "signal," which refers to the sentiment or tone of the news, generated using NLP. The second is "outcome," which comes from market data, reflecting what actually happens on that day. The main question is whether the "signal" aligns with the "outcome" or not, interpreting news sentiment as the signal and market return as the outcome.

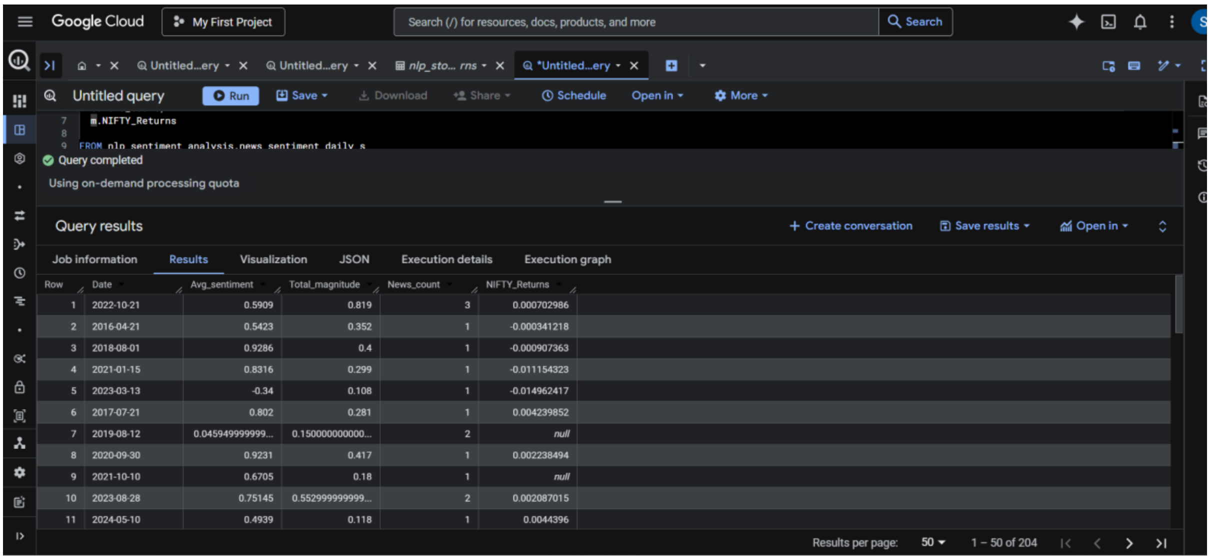
Since these signals contain news about the overall market, such as macroeconomic updates, views on future performance, major global updates, and the Indian economy, all of which influence market participants, participants use this information to decide whether to buy or sell, leading to price movements in specific stocks or indices.

So, the way I align the sentiment of news days with the market return value is through a left join. A left join takes everything from the primary table, which is on the left side, and matches it with the other table based on the common key (Date). I now have two different datasets: one for the market return and another one formed as an aggregated dataset. Both datasets are joined using a LEFT JOIN, with the primary dataset being the aggregated one.

As I mentioned, the challenge is news scarcity, so the main focus is on news days and observing their effect on the market return value using the common key (Date). This creates a dataset where both the signal and outcome are aligned for direct comparison.

After implementing the LEFT JOIN, there were some days when market returns were null, indicating that news days fell on weekends or holidays. Also, simply looking at each row's value is not enough to understand the relationship between the signal and outcome. So, the next question is how to perform proper analysis and truly understand the relationship between the signal and outcome.

In the end, this step focuses on aligning sentiment signals with actual market outcomes using a structured join, creating a comparable dataset that allows further analysis of whether news sentiment has any meaningful relationship with market behaviour.



Sub Question 4 -

There are two questions that I think I will focus on. The first is about analyzing the signal and output in a meaningful way, and the second concerns potential errors. Errors refer to the chances of failure and how the market sometimes behaves opposite to what the signal indicates. Understanding this is very important because there are times when, in the market, the signal suggests one thing but the actual market output turns out differently.

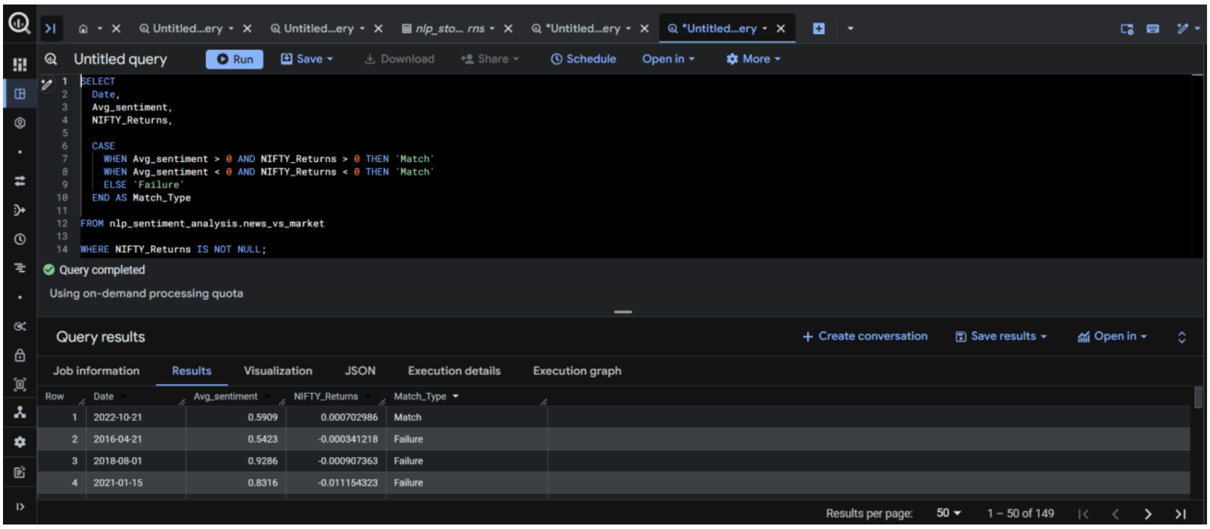
One of the reasons why this opposite behaviour occurs, which is commonly called a mismatch, is that market output depends on various factors. Sentiment is just one of these factors, and sometimes it cannot capture the overall tone of market participants. For example, there might already be an expectation of positive news that is priced into the market, or mixed signals could lead to confusion among participants.

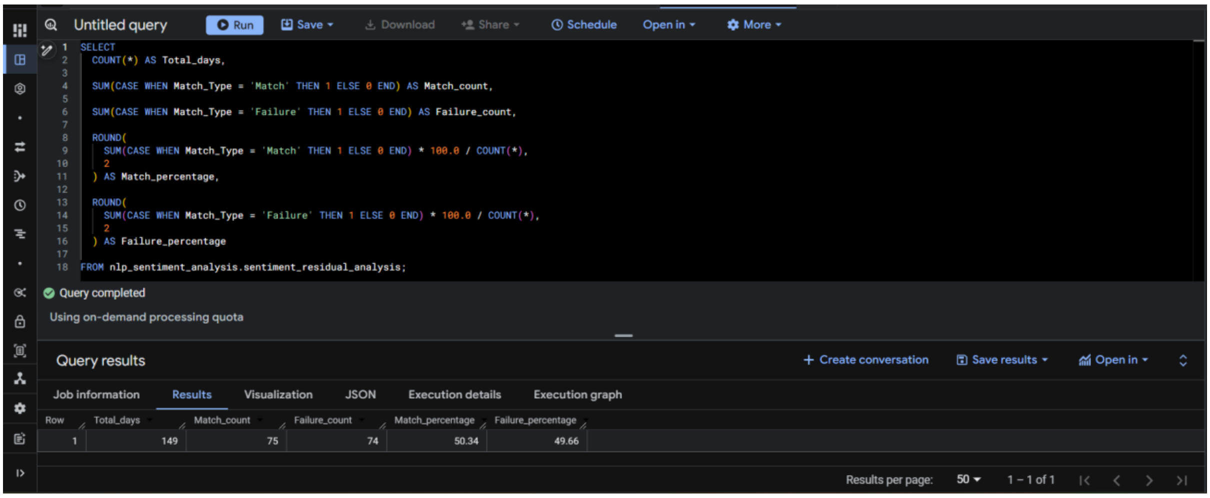
So, how do I perform the analysis and identify where the mismatch occurs? This will be done by creating a table of matches. Matches are where the signal aligns with the output, and mismatches are where it does not. This way, I can understand the overall picture of how much the signal aligns with the output and how much it fails. One important thing to keep in mind, which I have already highlighted, is the scarcity of the news that needs to be considered when analyzing the results.

The analysis does not provide a strong or consistent relationship, as the results show that nearly half the time mismatches occur, and the total days covered are only 149. When I compare this with the total market return days, which are 2467, it amounts to just about 6% of the days, likely due to the scarcity of news.

Returning to the analysis, if mismatches happen around half the time, it indicates that other factors also influence market behaviour, which suggests that sentiment alone does not have strong explanatory power for market movements. Sentiment is just one of the factors, and it also shows that sentiment does not consistently translate into market movement. This leads me to focus on finding other factors that may have a stronger influence on market behaviour.

So, overall, since nearly half of the outcomes result in mismatches, it can be said that sentiment is a weak signal for understanding market behaviour, especially under conditions of limited data, and it cannot be relied upon as a standalone factor.





Answer of the First Major Question -

The first major question focuses on the sentiments generated by financial news, which offer valuable insights into market movement. The idea was to understand whether these sentiments can act as a meaningful signal to explain how the market behaves on specific news days.

From the analysis that I have done, the answer seems to be that sentiment is a weak signal, as nearly half of the time it was opposite to the actual market movement. This shows that even though sentiment captures the tone of the news, it does not consistently translate into market outcomes. There are many instances where the signal suggests one direction, but the market behaves differently.

One of the main reasons for this is that market behaviour depends on multiple factors, and sentiment is just one of them. News sentiment reflects how information is presented, but the market may already have expectations priced in, or there may be other macroeconomic or global factors influencing the outcome. Because of this, sentiment alone is not able to capture the full picture of how market participants react.

Another important limitation comes from the data itself. Since the analysis is based on limited news days due to data scarcity, the sample size becomes small compared to the total market days. This further reduces the reliability of the signal, as it does not cover enough instances to build a strong relationship.

So, overall, the result shows that sentiment does not provide a strong or consistent signal for understanding market behaviour. It can give some level of context, but it cannot be relied upon as a standalone factor. This also indicates that understanding market behaviour or movement is not easy with just one signal like sentiment, and it requires exploring other factors that may have a stronger influence.

Second Major Question of the Project -

After conducting the sentiment analysis and reviewing the results, one thing is clear: sentiment itself is a weak signal, as the mismatch rate was nearly 50%. This highlights that many other factors influence market movement or behaviour.

Sentiment focuses only on the tone of the news, not on the actual content or the subjects being discussed. This raises another doubt—many news items may carry positive sentiment but may not be important enough to influence the market. In reality, the market reacts more to the underlying factors involving companies, events, and macroeconomic conditions rather than just the tone.

This leads to the second major question: what if I shift the focus from sentiment to the entities mentioned in financial news? Entities represent the main subjects of the news, such as companies, events, or macroeconomic terms.

The question then becomes whether focusing on these entities, instead of just the tone of the news, can provide a more meaningful and stronger signal for understanding market movement or behaviour.

Overall, this question shifts the focus from how the news sounds to what the news is actually about, aiming to test whether the core subjects of the news carry more meaningful information than sentiment alone in explaining market behaviour.

Sub Question 1 -

So, if I will be using entities to understand market behaviour, the first question is how I extract them from the news data, and what numerical values I am going to focus on while extracting these entities. There is another point I have already mentioned, that the platform or tool I am using for analysis, which is BigQuery SQL, does not directly support NLP functions. So, I will be using Python to extract the functions that I need for entities, and after extracting those numerical values, I will perform the analysis in BigQuery SQL itself.

So, what is the meaning of the entity that I am going to extract? In very simple terms, an entity is the important element in a sentence, the "who" or "what" in the sentence, like a company name, an event name, or any macroeconomic term. It is the key component or group of components in a sentence. In other words, entities are the drivers of the sentence.

So now, how am I going to do that, and what NLP functions am I going to use? For this, there are two concepts that need to be understood. First is NER, or Named Entity Recognition, which reads the text and identifies entities, assigning them to different labels like names, places, organizations, etc. It is like our brain, which naturally identifies the most important elements in a sentence. In NLP, that is done by NER.

The other important function is the salience score, also known as the importance score. This score focuses on the entities in a sentence and ranks them according to the importance they carry within that sentence. So, in simple terms, it means ranking the entities sentence-wise based on their relative importance.

These two functions convert the news data into entity-based structured data that is needed for this question to be analyzed.

There is another challenge I am going to face while using Python, as the Python NLP library does not have a salience score by default, so I am going to calculate it myself. The two rules I am going to use for the salience score are the position of the entity in the sentence (earlier position means a higher salience score) and a ranking based on the entity types present in the sentence (for example, giving a company entity type a higher ranking). I will combine both to create a salience score for each entity, so both position and entity type matter for the final score.

In Python, I am using the spaCy NLP model, which is used to process text or unstructured data into structured data and includes the NER function, so it is an entity-based NLP model. In spaCy, NER has entity types such as ORG for company names, GPE for locations, and many more. For the salience score, I have assigned 1 to ORG, 0.7 to GPE, and 0.3 to other entity types. The position score is based on the reciprocal of the entity's position; for example, if an entity is at position 2, the position score is 1/2. Multiplying the entity type score with the position score gives the final salience score for that entity.

This approach provides a simple and interpretable way to measure entity importance in short news text, but it does not capture deeper contextual or semantic importance.

After creating these entity functions, I have uploaded them into BigQuery SQL, as further analysis will be performed there.

In the end, this step focuses on converting unstructured text into entity-based structured signals by identifying key subjects in the news and assigning them measurable importance, creating a foundation for analyzing whether these entities provide a stronger signal than sentiment.

Date	News	Entity	Entity_Type	Salience
14-02-2015	HDFC Bank reported quarterly net profit growth of around 2	HDFC Bank	ORG	1
14-02-2015	HDFC Bank reported quarterly net profit growth of around 2	quarterly	DATE	0.15
14-02-2015	HDFC Bank reported quarterly net profit growth of around 2	around 20%	PERCENT	0.1
14-02-2015	HDFC Bank reported quarterly net profit growth of around 2	Reuters	ORG	0.25
21-04-2016	HDFC Bank announced strong quarterly earnings with stabl	HDFC Bank	ORG	1
21-04-2016	HDFC Bank announced strong quarterly earnings with stabl	quarterly	DATE	0.15
21-04-2016	HDFC Bank announced strong quarterly earnings with stabl	NPAs	GPE	0.233333333
21-04-2016	HDFC Bank announced strong quarterly earnings with stabl	Indian	NORP	0.075
08-11-2016	Following demonetisation announced by the Indian governn	Indian	NORP	0.3
08-11-2016	Following demonetisation announced by the Indian governn	HDFC Bank	ORG	0.5
08-11-2016	Following demonetisation announced by the Indian governn	Wikipedia	ORG	0.333333333
13-01-2017	HDFC Bank reported steady growth driven by expansion in r	HDFC Bank	ORG	1
17-07-2017	HDFC Bank delivered consistent quarterly performance wit	HDFC Bank	ORG	1
17-07-2017	HDFC Bank delivered consistent quarterly performance wit	quarterly	DATE	0.15
20-04-2018	HDFC Bank posted strong earnings growth with stable asse	HDFC Bank	ORG	1
01-08-2018	HDFC Bank became Indiaâ€™s most valuable lender by mar	HDFC Bank	ORG	1
01-08-2018	HDFC Bank became Indiaâ€™s most valuable lender by mar	India	GPE	0.35
14-01-2019	HDFC Bank reported strong quarterly earnings supported b	HDFC Bank	ORG	1
14-01-2019	HDFC Bank reported strong quarterly earnings supported b	quarterly	DATE	0.15
20-04-2019	HDFC Bank maintained steady profit growth and stable mar	HDFC Bank	ORG	1
21-10-2019	HDFC Bank continued expanding its digital banking service	HDFC Bank	ORG	1
27-03-2020	HDFC Bank announced support measures during COVID-1	HDFC Bank	ORG	1
27-03-2020	HDFC Bank announced support measures during COVID-1	COVID-19	ORG	0.5
18-07-2020	Despite pandemic-related disruptions, HDFC Bank reporte	HDFC Bank	ORG	1
02-12-2020	RBI barred HDFC Bank from issuing new credit cards after r	RBI	ORG	1
02-12-2020	RBI barred HDFC Bank from issuing new credit cards after r	HDFC Bank	ORG	0.5

news_entity_base

Sub Question 2 -

After extracting entities from news data, another question arises. Many of the entities I have extracted may not be very important for market behaviour. I have already labeled the entities with salience scores, so the new question is which entities matter most for analyzing market behaviour. If I do not include the important entities, the analysis will also be affected by the noise that many entities can create.

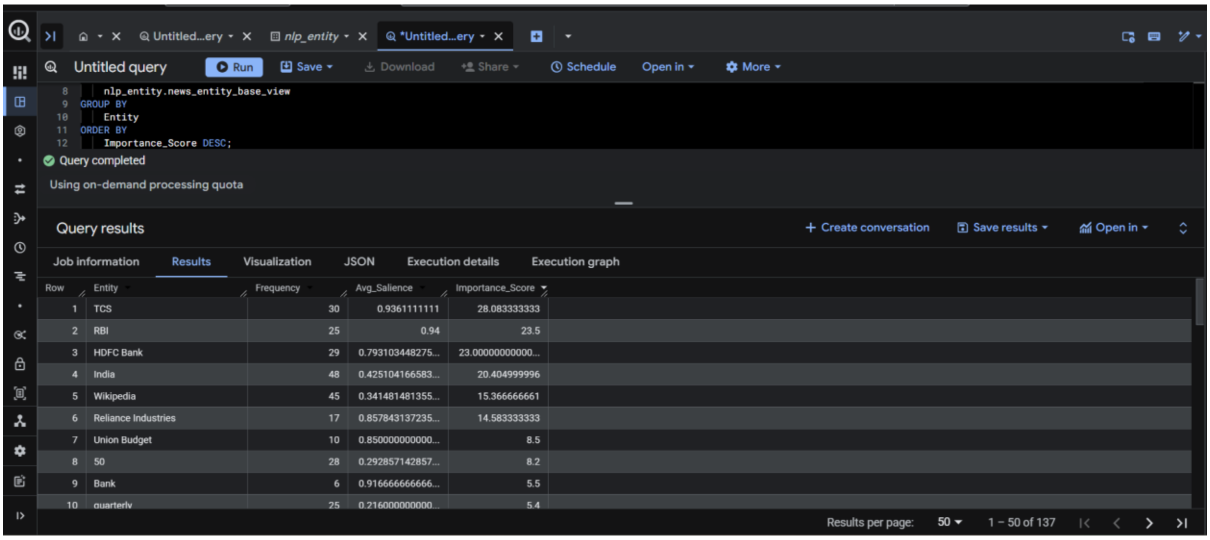
But there is another challenge that I have understood is always present in NLP analysis, and that is the bias in the news data itself. If the news data is biased toward certain companies, the analysis will always be poor. This is different from market data, which is easily available and well maintained. News data is not as reliable, so the dataset itself can make the whole analysis uncertain in many ways. This becomes a clear limitation in NLP analysis, and it is even more pronounced with older news data, like the 10-year dataset I am using.

Now, coming back to the question of how I can figure out which entities matter more than others for analyzing market behaviour, I will do that using two rules. The first is the frequency with which an entity appears in the overall entity data that I extract, and the second is the average salience of that entity. Sometimes noise or less important items can appear many times as supporting entities, but they do not have a high salience score. By categorizing in both ways, it becomes easier to understand which entities are important for further analysis.

I am going to multiply these two to get a single score that defines which entities are more important for further analysis. When I implemented this and identified the entities that are more important than others, the result itself shows that some words like “quarterly” appear many times, but they are not that salient. Another example is the word “Wikipedia,” which comes from the source of the news. This also shows that the extraction of important entities depends on how the news data is collected.

Now the next question comes up: how to use these entities, which I see as important, to analyze market movement.

So, overall, this step focuses on filtering and identifying the most relevant entities by reducing noise and accounting for data bias, so that only meaningful subjects are used for further analysis of market behaviour.



Sub Question 3 -

After understanding and extracting the important entities, the main question is whether these entities align with market behaviour or not. In other words, the goal is to test the relationship between the entity signals and market movement.

So how am I going to do it? The approach is again signal- and outcome-based, as I have used before. Now the entities become the signal, and the market returns become the outcome. So, is there any impact or any useful understanding when entities are used as the signal and market returns are the outcome?

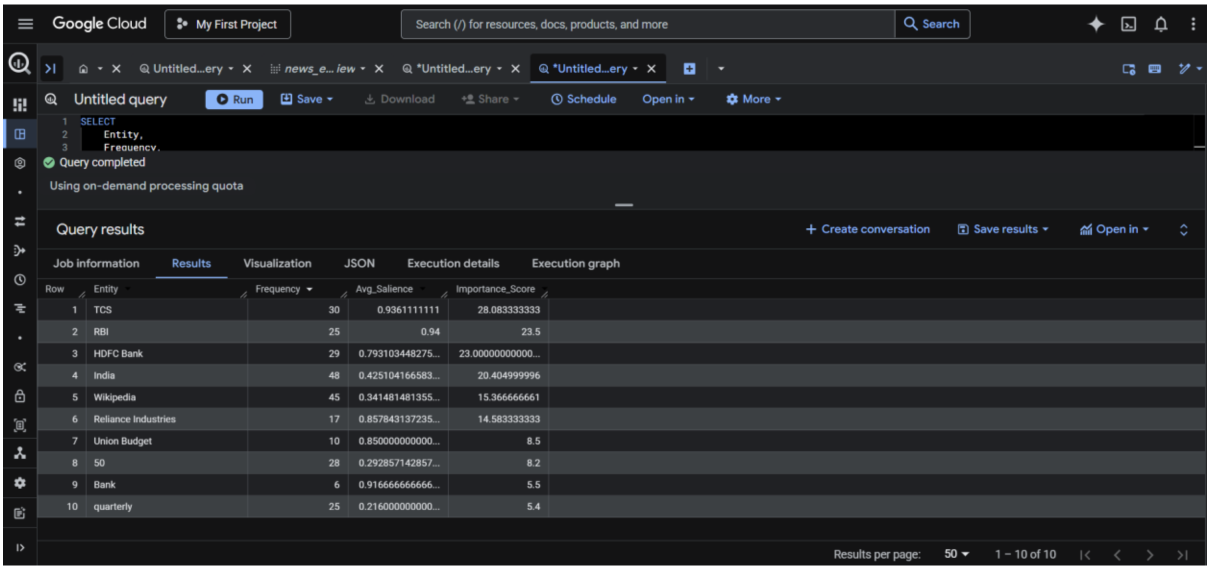
There are some key points to understand. Because there are many entities for each date, alignment will be achieved by joining each entity by date. I need to understand the overall pattern of an entity. If an entity appears frequently in the dataset, does it signal market returns in the same way or not? That would reveal a pattern. If an entity appears many times but does not indicate a specific pattern, then another factor may be influencing market movement, and the entity may just be noise.

Now, the implementation goes as follows. As I have already calculated entity importance, I am selecting only those entities with an importance score greater than 5. As a result, only 10 entities come up out of the 137 entities that were there in the entity importance. This also shows that many entities were insignificant, appeared very little, or had incomplete types.

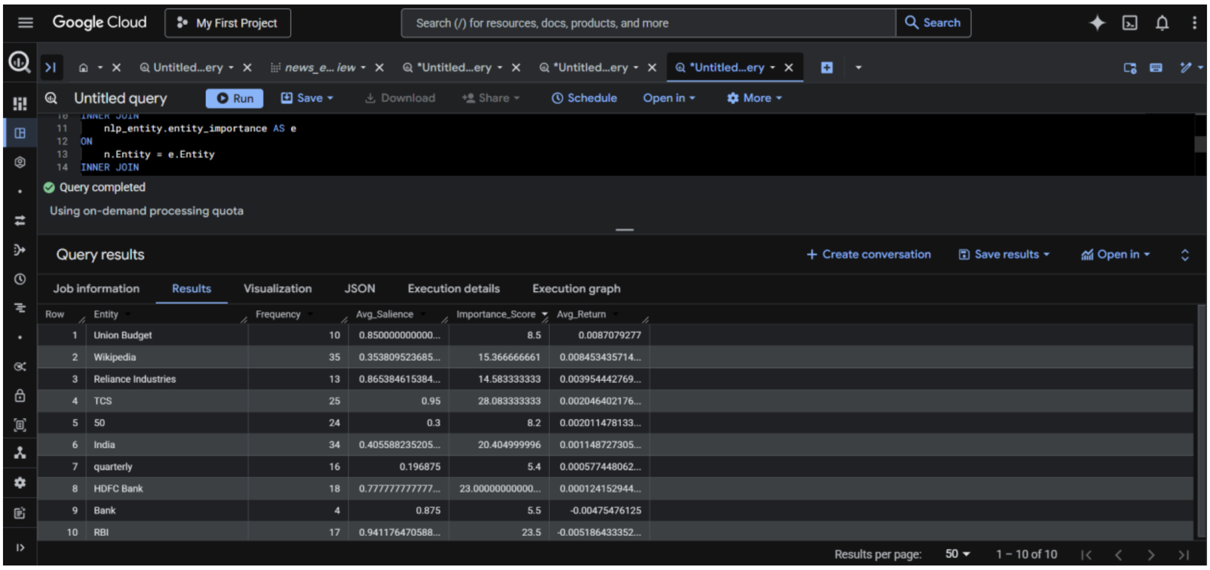
What I have done next is calculate the average NIFTY returns with respect to these entities, which helps to see whether each entity shows a clear pattern or not.

The result is slightly surprising. There are only 10 entities, and while the Union Budget shows a higher average market return, it has a relatively low frequency, which means the signal may be event-driven but not very stable. On the other hand, entities like HDFC Bank and TCS, which have higher salience and frequency, do not show a strong or consistent average return. This suggests that these entities are associated with both positive and negative market movements, which cancels out the overall effect.

So overall, most of the entities do not show a strong or consistent alignment with market returns. This indicates that even after filtering by importance, many entities may still behave like noise in terms of market understanding, and only a few event-based entities may show some level of impact, but not consistently enough to rely on directly. In the end, the analysis shows that focusing on entities improves understanding compared to sentiment, but most entities still do not provide a stable or reliable signal, indicating that entity-based signals alone are not sufficient to explain market behaviour consistently.



Row	Entity	Frequency	Avg_Salience	Importance_Score
1	TCS	30	0.9361111111	28.083333333
2	RBI	25	0.94	23.5
3	HDFC Bank	29	0.793103448275...	23.00000000000...
4	India	48	0.425104166583...	20.404999996
5	Wikipedia	45	0.341481481355...	15.366666661
6	Reliance Industries	17	0.857843137235...	14.583333333
7	Union Budget	10	0.850000000000...	8.5
8	50	28	0.292857142857...	8.2
9	Bank	6	0.916666666666...	5.5
10	quarterly	25	0.216000000000...	5.4



Row	Entity	Frequency	Avg_Salience	Importance_Score	Avg_Return
1	Union Budget	10	0.850000000000...	8.5	0.0087079277
2	Wikipedia	35	0.353809523685...	15.366666661	0.008453435714...
3	Reliance Industries	13	0.865384615384...	14.583333333	0.003954442769...
4	TCS	25	0.95	28.083333333	0.002046402176...
5	50	24	0.3	8.2	0.002011478133...
6	India	34	0.405588235205...	20.404999996	0.001148727305...
7	quarterly	16	0.196875	5.4	0.000577448062...
8	HDFC Bank	18	0.777777777777...	23.00000000000...	0.000124152944...
9	Bank	4	0.875	5.5	-0.00475476125
10	RBI	17	0.941176470588...	23.5	-0.005186433352...

Sub Question 4 -

The results I have obtained from the important entities in relation to market movements were not very strong. Still, to understand the signals that are somewhat important, even if they are not as significant as expected, filtering these entities with respect to high and low signals should be done. Even though there is not much real progress from sentiment to entity, in my opinion, this step helps in understanding the relative impact better.

The main reason, as I believe, is the scarcity of the news database. I am working with a dataset covering over 10 years, and finding that much news is challenging, and there can also be bias in how it is collected, as I mentioned before. But I think if the market data were taken for a shorter period, like 2–3 months, there would be ample news available for that timeframe, which could reduce the challenge of scarcity, although bias in data collection could still remain.

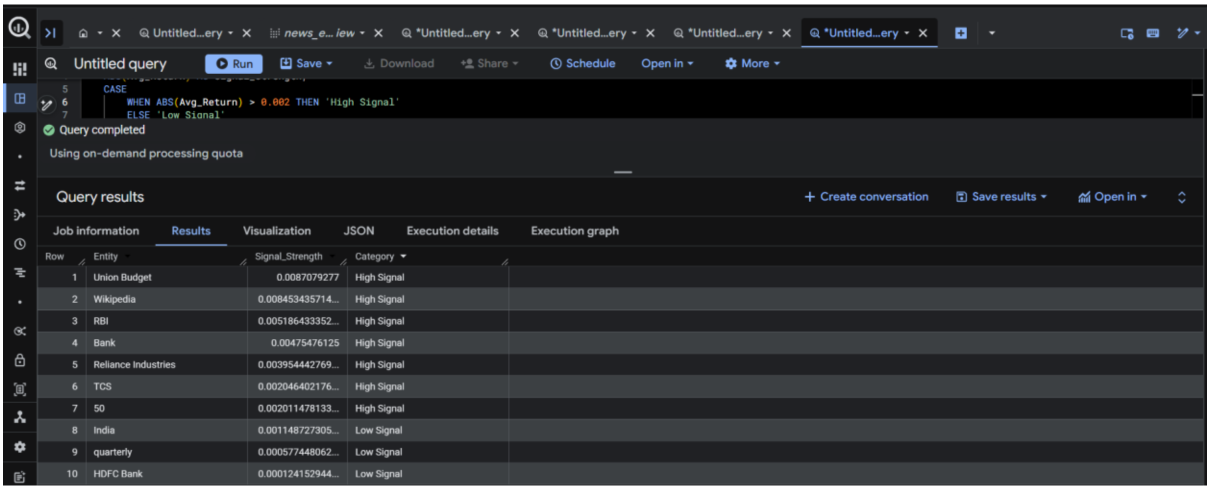
So how am I going to filter the entities into strong and weak signals? Here, strong and weak signals refer to the impact these entities have on market movement. I can see some entities with high salience and frequency, like TCS and HDFC, that show both positive and negative returns, which leads to inconsistency. But an entity like the Union Budget does not appear as often, and whenever it does, it tends to show stronger movement in market returns, though the number of observations is limited. So the filtering is done to understand market movement, not just based on how an entity has been characterized by salience and frequency.

So I am going to use a rule to differentiate between strong and weak signals. It will be as follows: if the absolute average Nifty return is greater than 0.2%, it will be considered a strong signal; otherwise, it will be a weak signal. Taking the absolute value also matters because some entities can consistently pull Nifty returns in the negative direction.

The results did not change in terms of numbers, but this step categorizes them as high/low or strong/weak signals. The main observation is that words like “Wikipedia” and “bank” appear as high signals, even though they do not have much salience and usually appear later in the news as supporting entities. This shows that the classification is capturing the magnitude of movement, but not necessarily the reliability or true importance of the entity.

Entities like RBI and Union Budget seem to be more relevant, but the Union Budget is more event-based. RBI's monetary policy can be seen as having a major effect on the market, though in my data it appears more on the negative side when looking at average Nifty returns.

Overall, there is not much strong evidence from the entity analysis that clearly defines entities as a consistent driving force behind market movement, which suggests that entity-based signals alone are not sufficient to explain market behaviour.



Row	Entity	Signal_Strength	Category
1	Union Budget	0.0087079277	High Signal
2	Wikipedia	0.008453435714...	High Signal
3	RBI	0.005186433352...	High Signal
4	Bank	0.00475476125	High Signal
5	Reliance Industries	0.003954442769...	High Signal
6	TCS	0.002046402176...	High Signal
7	50	0.002011478133...	High Signal
8	India	0.001148727305...	Low Signal
9	quarterly	0.000577448062...	Low Signal
10	HDFC Bank	0.000124152944...	Low Signal

Answer of the Second Major Question -

Before answering the question, there is something important that I understand: analysing NLP-based data is much more dependent on the dataset itself. The main challenge I have faced is the scarcity of the news, which I had already identified earlier. But another factor that becomes important in entity analysis is how the news is collected, as that can introduce bias into the dataset.

When I compare this with the ML analysis I did in the previous project, the scarcity of market return data and bias in returns was not a problem, as the data was historically available and purely quantitative. But here, I am working with qualitative data, which brings more challenges. These two challenges—scarcity of news and bias in data collection—are clearly reflected in the entity analysis as well.

As a result, the outcome was not strong enough to conclude that entities can act as major factors influencing market behaviour, or that they have a clear and consistent impact on market movement.

The results suggest that entity-based signals alone are not strong enough to consistently explain market movement, as many entities either behave inconsistently or act as noise, even after applying importance and signal filtering. Even entities that appear frequently with higher salience do not show a stable pattern, while some event-based entities show movement but lack consistency due to limited observations.

This shows that entity analysis may provide some level of contextual understanding, especially in identifying what the news is about, but it cannot be relied upon as a standalone driver for market behaviour.

Another important point is that the results can change if the dataset used is more recent, such as a 3–4 month period, where the scarcity of news would not be a major issue. In such cases, the signal strength could improve due to better data coverage. However, the bias in how the news is collected would still remain a challenge, and that can still affect the reliability of the analysis.

In the end, the analysis shows that shifting from sentiment to entities improves the depth of understanding by focusing on what the news is about, but it still does not provide a strong or reliable signal on its own, mainly due to data limitations and the inherent noise present in entity-based extraction.

Third Major Question of the Project -

After realizing and understanding the limitations I am facing, mainly the bias in how the data is being collected, I have been deeply concerned about it. The scarcity of news can be addressed by using more recent data, but how can this bias, which can arise naturally during data collection, be handled? When working with entity names, everyone can have some preferences with regard to companies, institutional bodies, and other factors, so they may tend to collect those types of news more for analysis.

I think this limitation can be reduced, although I am still not completely sure. From my point of view, the market is driven more by themes that are present in the world, rather than by specific entities. For example, if there is a war that leads to an economic slowdown, the war itself becomes one theme, the slowdown becomes another, and a recession becomes another. Similarly, the formation of a bubble can also be considered a theme. I am trying to understand what these themes are like in nature and how they influence the market.

In these themes, entities may not matter as much, as the theme captures the broader market movement. This can also help reduce bias in how the news is collected, since most major entities are influenced by the overall theme rather than isolated events. In my opinion, this may not be entirely correct, but it is something I want to test.

So, the next major question is whether themes can be useful for understanding market behaviour or not. This question shifts the focus from individual entities to broader market narratives, aiming to test whether themes can provide a more stable and less biased signal for understanding market behaviour.

Sub Question 1 -

So, the first thing is that themes are normally a group of patterns that indicate a particular result or a particular thing, which is called a theme. So what are these patterns, and from where do they extract a common theme? These patterns are called phrases, which are the meaningful output in a sentence.

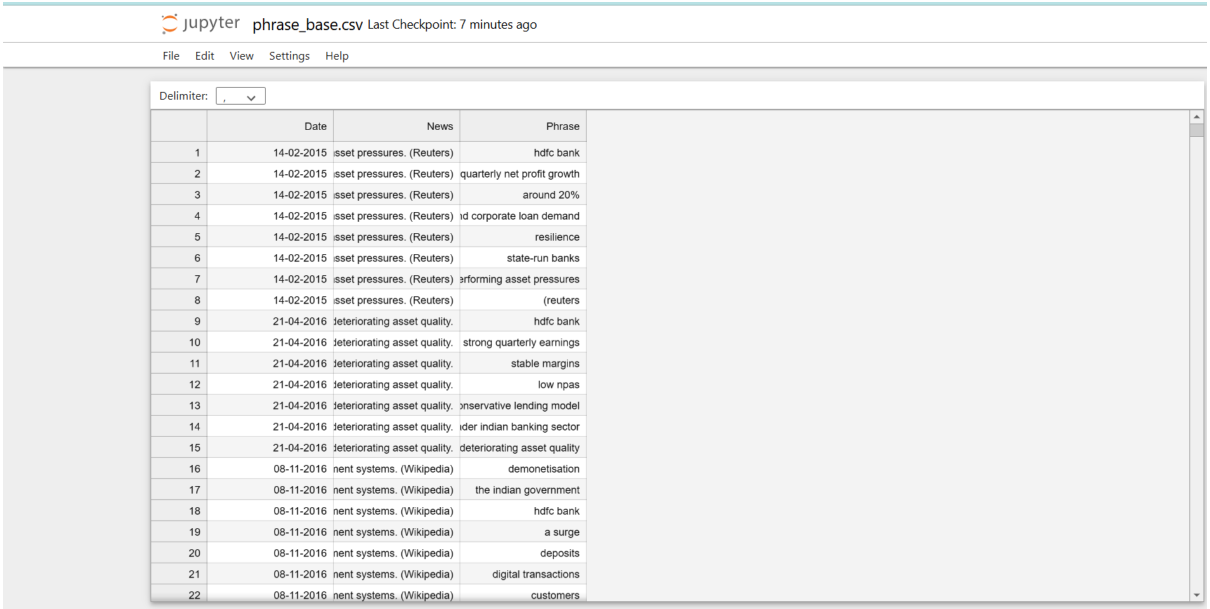
Normally, in entity analysis, I was focused on the “who” part, but now the main focus is on the meaningful situations or events present in the sentence. In simpler terms, the things that are happening or the results that are being conveyed are the phrases. There can be more technical definitions, but in very simple terms, it is a meaningful output or a group of words in a sentence that gives a clear meaning.

So, the main question is how I am going to extract these phrases from the news data. As I have done with sentiment and entities, this will also be done using NLP functions. I have already explained NLP, which is converting raw text into structured, meaningful data that can be understood by machines for processing. Since BigQuery SQL does not support direct NLP functions, I will be doing this phrase extraction using Python and then performing further analysis on it in BigQuery SQL, similar to what I have done so far in this project.

Now, for extracting phrases, I am using the spaCy library in Python, which I have already used for entities. This time, I am focusing on key phrases. In Python, I am using the function called noun chunks. It identifies continuous groups of words around a noun that form meaningful input or output. It mostly captures noun-based meaningful phrases rather than action verbs, which I have already handled in entities. These are the phrases I am focusing on.

I created the dataset with three columns: the first is date, the second is news, and the third is the phrases. Each row contains a single phrase, so multiple rows can have the same date and news, as one news item can have multiple phrases. After that, I uploaded this into BigQuery SQL, and further analysis will be done there.

Overall, this step focuses on extracting meaningful phrases from the news to capture broader situations or events, forming the base for identifying themes that may better explain market behaviour compared to sentiment and entities.


jupyter phrase_base.csv Last Checkpoint: 7 minutes ago

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	Date	News	Phrase
1	14-02-2015	sset pressures. (Reuters)	hdfc bank
2	14-02-2015	sset pressures. (Reuters)	quarterly net profit growth
3	14-02-2015	sset pressures. (Reuters)	around 20%
4	14-02-2015	sset pressures. (Reuters)	d corporate loan demand
5	14-02-2015	sset pressures. (Reuters)	resilience
6	14-02-2015	sset pressures. (Reuters)	state-run banks
7	14-02-2015	sset pressures. (Reuters)	erforming asset pressures
8	14-02-2015	sset pressures. (Reuters)	(reuters
9	21-04-2016	deteriorating asset quality.	hdfc bank
10	21-04-2016	deteriorating asset quality.	strong quarterly earnings
11	21-04-2016	deteriorating asset quality.	stable margins
12	21-04-2016	deteriorating asset quality.	low npas
13	21-04-2016	deteriorating asset quality.	onservative lending model
14	21-04-2016	deteriorating asset quality.	der indian banking sector
15	21-04-2016	deteriorating asset quality.	deteriorating asset quality
16	08-11-2016	nent systems. (Wikipedia)	demonetisation
17	08-11-2016	nent systems. (Wikipedia)	the indian government
18	08-11-2016	nent systems. (Wikipedia)	hdfc bank
19	08-11-2016	nent systems. (Wikipedia)	a surge
20	08-11-2016	nent systems. (Wikipedia)	deposits
21	08-11-2016	nent systems. (Wikipedia)	digital transactions
22	08-11-2016	nent systems. (Wikipedia)	customers

Sub Question 2 -

So, now that the phrases have been extracted, how do I determine which phrases are more important than others and how to separate them? This is now the main question.

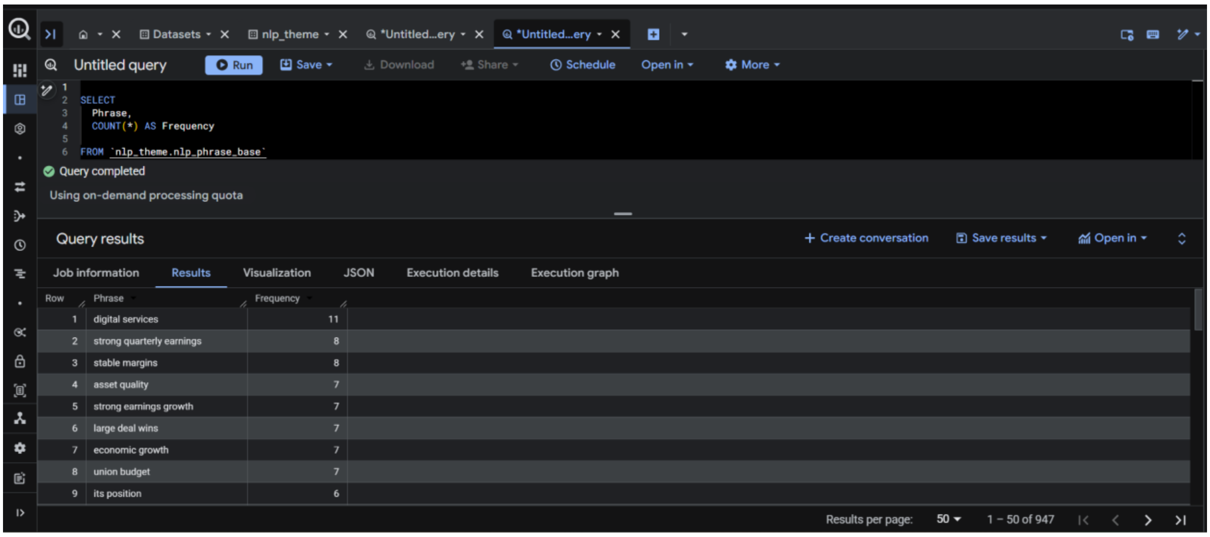
The answer to this is the frequency of these phrases. A phrase that appears more often seems to indicate a theme that many news sources are covering. Phrases that appear only once or twice can be considered less reliable or weaker signals, since they may not be important enough to represent a broader theme.

When I started implementing phrase frequency using count in BigQuery SQL, words like "Wikipedia," "HDFC Bank," and others appeared here as well and had the highest frequency, which created a challenge. In the phrase extraction step, I did not initially realize that noun chunks would include both single nouns and multi-word phrases.

So, I filtered out single words and other terms that I considered not useful as phrases, such as company names, because the focus is on meaningful units or outputs, as these patterns or phrases will eventually form part of broader themes.

This step also highlights another challenge: extracting meaning from text is more biased compared to market return data, as text data is inherently subjective and depends heavily on interpretation and filtering decisions.

In the end, this step focuses on identifying meaningful and frequently occurring phrases to represent potential themes, while also handling noise and subjectivity that naturally arise in text-based analysis.



Sub Question 3 -

After determining the frequency of the phrases, the next step is to group them into themes. The main question is how to group these patterns into themes. A theme is one of the important indicators of market movement because it contains many patterns that work together in a chain reaction. In simpler terms, a group of similar phrases that represent the same meaning is called a theme.

I understand that there are too many small signals and too many entities. If I focus on a single signal alone, it will be weak, because the market often moves by combining multiple signals into one, which, in my view, works like a chain reaction. That entire chain reaction reflects one theme, which becomes a stronger signal.

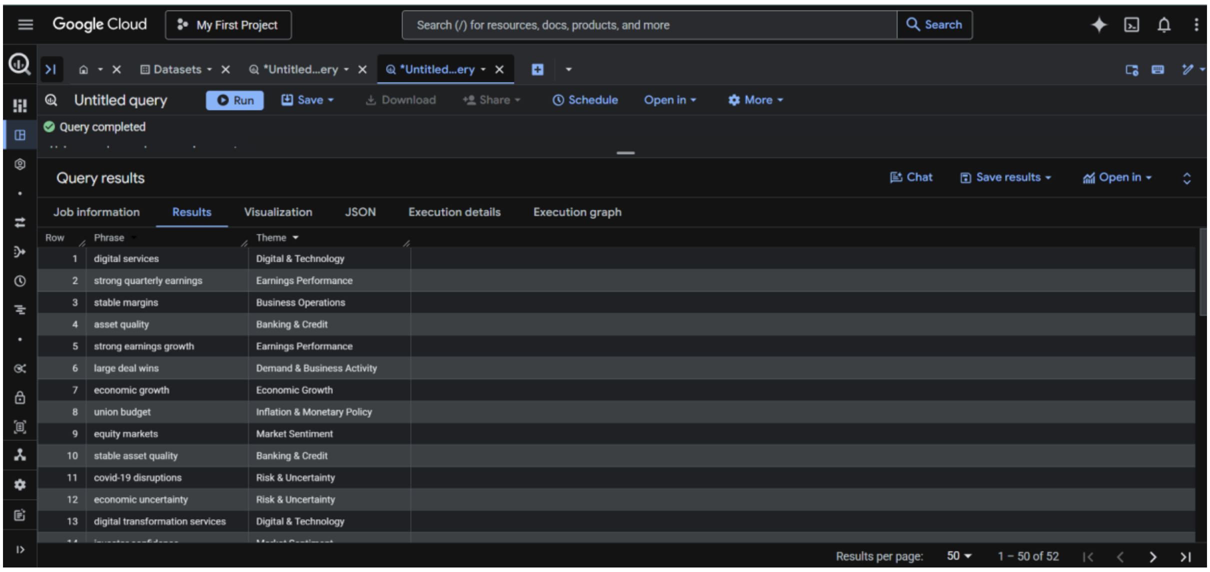
For example, a recession is a main theme. Inside it are job losses, decreases in savings, decreases in spending capacity, lower wages, and unemployment. These smaller signals combine into a bigger theme, which is recession, and this theme helps explain how the market behaves.

So, how do I create themes from these phrases? I am using manual grouping, in the same way I explained in the recession example. I look at what the phrases represent at a broader level, and based on that understanding, I group them into themes.

When I started implementing manual grouping, there were many challenges. First, many phrases appeared only 1–2 times, which created noise, so I filtered them by frequency, keeping only those that appeared 3 or more times. After this filtering, I had 52 phrases, which I grouped into 8–9 clusters where it seemed appropriate.

This process is highly subjective, so the results depend on how the manual grouping is done. After grouping, I assigned these clusters as themes that I thought they belonged to, and mapped these themes to the phrases.

Overall, this step shows that themes can capture broader market narratives by combining multiple smaller signals, but the process of creating these themes is subjective and depends heavily on interpretation, which can influence the final analysis.



Sub Question 4 -

Now after extracting the themes, the obvious question is whether these themes can show a meaningful impact on how the market behaves. Is there any strong signal that can be observed through these themes in relation to market behaviour or not? Can themes be useful for understanding market behaviour or not?

So now, as I have done before, the themes become the signal and the market return becomes the outcome. The main focus is to check whether there is any consistent impact on the outcome with respect to these signals. This also depends on how I have done the manual grouping, as NLP-based analysis is more subjective compared to ML. In ML, we have numerical data and well-defined analysis methods, whereas in NLP, working with text involves subjectivity in how the data is defined, collected, and filtered.

In the implementation, the process has multiple steps. First, there was no date column in the theme view that I had created earlier, so I added that column, as the date becomes the main joining key between the theme data and the market return data. After joining the market return data with the themes, I observed that the returns were at a single-phrase level, which is not what I was aiming for.

To address this, I used the frequency of theme occurrences along with the average Nifty return, so that I could understand both the presence of the signal and its impact on the market. This helped me identify some themes that show meaningful relationships with market movement.

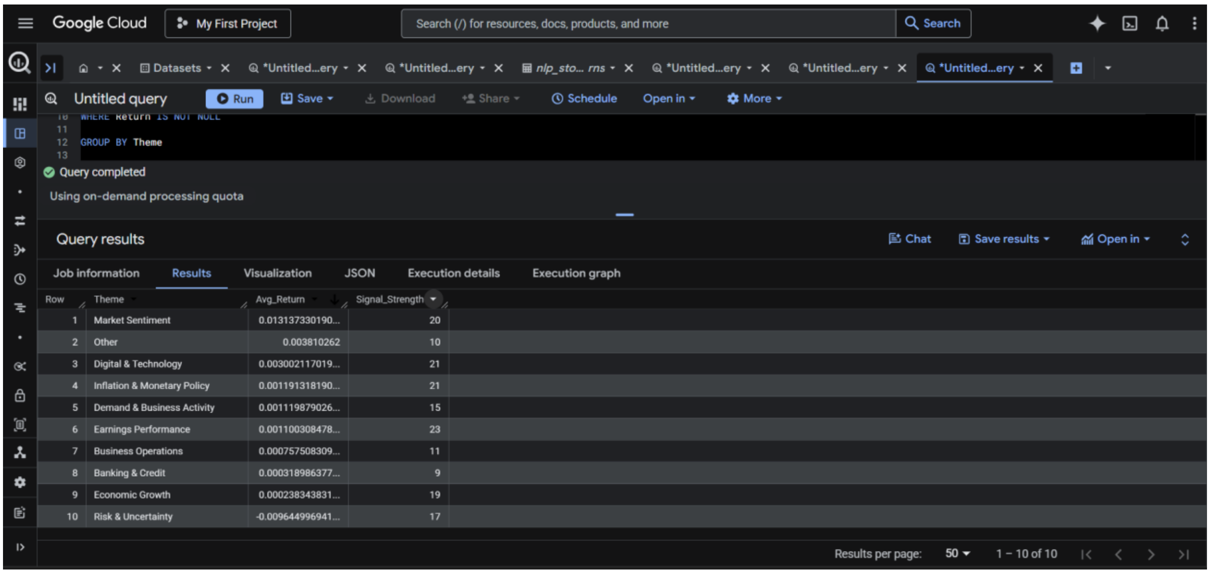
One of the major themes is market sentiment, which includes phrases like “strong investor confidence,” “investor sentiment,” and “record high.” This indicates that when such news appears, the market reacts positively, with an average Nifty return of around 1.3%, showing a strong relationship.

Another theme is digital and technology, which includes phrases like “digital services growth” and “digital transformation.” This shows that the market reacts positively to developments in technology and digital growth, with an average Nifty return of around 0.3%.

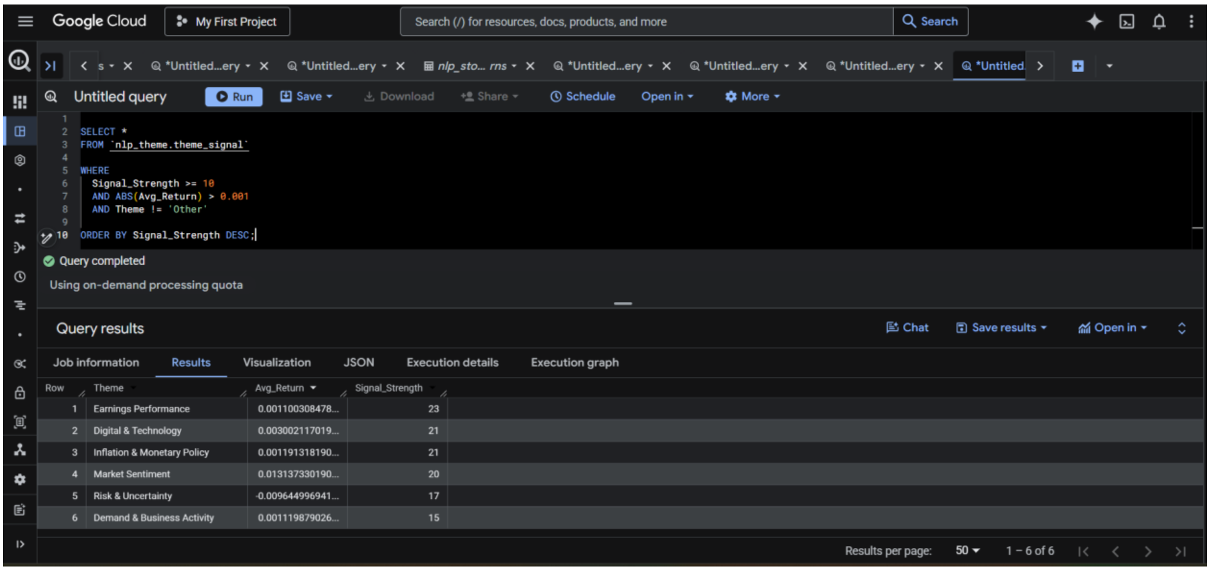
Then there is the risk and uncertainty theme, which has a negative impact on the market. This includes phrases like “COVID-19 disruption,” “regulatory scrutiny,” and “governance concerns.” These reflect the impact of the pandemic, regulatory pressure, and internal management issues, which negatively affect the market, with an average Nifty return of around -0.96%.

There are other themes as well, but they show weaker or inconsistent impact on the market, as they do not consistently influence market behaviour in a clear way.

Overall, I can say that themes provide a more meaningful understanding of market behaviour compared to sentiment and entity analysis, as they capture broader patterns that are closer to how the market actually reacts. However, the limitation of news scarcity still exists due to the 10-year dataset, although the bias seen in entity-based analysis is reduced when using themes.



Row	Theme	Avg_Return	Signal_Strength
1	Market Sentiment	0.013137330190...	20
2	Other	0.003810262	10
3	Digital & Technology	0.003002117019...	21
4	Inflation & Monetary Policy	0.001191318190...	21
5	Demand & Business Activity	0.001119879026...	15
6	Earnings Performance	0.001100308478...	23
7	Business Operations	0.000757508309...	11
8	Banking & Credit	0.000318986377...	9
9	Economic Growth	0.000238343831...	19
10	Risk & Uncertainty	-0.009644996941...	17



Answer of the third major question -

Two challenges or limitations emerged as I progressed in this project. The first was the scarcity of news data due to the 10-year period, and the second, which was more challenging, was the way the data was collected, which introduced bias toward the entity analysis, as I have already mentioned with companies and other factors. These limitations made it difficult to rely on sentiment and entity-based signals, as both were affected either by lack of sufficient data or by the way the data was filtered and selected.

So, when I moved to theme analysis, the results were much better compared to both sentiment and entity analysis. There were clearer and stronger signals visible in the theme analysis, both in positive and negative directions. These results were also understandable from a practical point of view, as the themes reflected broader situations that actually influence how the market behaves. While this understanding can be intuitive if someone thinks deeply about how the market reacts, having numerical evidence supporting it makes the conclusion more reliable.

One important observation here is that themes capture the broader narrative behind the news rather than focusing on isolated elements. Sentiment focuses only on tone, and entities focus on specific subjects, but themes combine multiple related signals into one structure. This makes the signal stronger, as it reflects how different factors interact together, which is closer to how the market actually reacts in real life.

For example, themes like market sentiment, digital growth, and risk or uncertainty show consistent behaviour in the market. Positive themes tend to align with positive market movement, while risk-related themes tend to align with negative movement. This level of alignment was not clearly visible in the earlier analyses, which shows that themes are able to capture a more meaningful relationship with market behaviour.

Another important point is that theme-based analysis helps reduce the bias that was present in entity-based analysis. Since themes are formed by grouping multiple phrases and patterns, they are less dependent on specific companies or entities, and more focused on the overall situation. This reduces the impact of how the news is collected and shifts the focus toward broader market narratives.

However, the limitation of data scarcity still exists. Even though themes provide a better signal, the number of observations is still limited due to the dataset covering a long historical period with fewer available news items. This means that while the signal is stronger, it is still not fully reliable in all situations.

In the end, the results show that themes provide a more meaningful and structured way to understand market behaviour compared to sentiment and entity-based approaches, as they capture broader narratives that align more closely with how the market actually reacts, even though limitations in data availability still remain.

Fourth Major Question of the Project -

After analyzing sentiment, entity, and then themes, themes have shown relatively stronger signals for understanding market behaviour. But there is one thing I have been thinking about. If sentiment and entity, when taken alone, are seen as weak signals, what if I combine all three of them?

Will they complement each other and produce a stronger and more complete signal, or are these weak signals just noise that could negatively impact the theme-based signal?

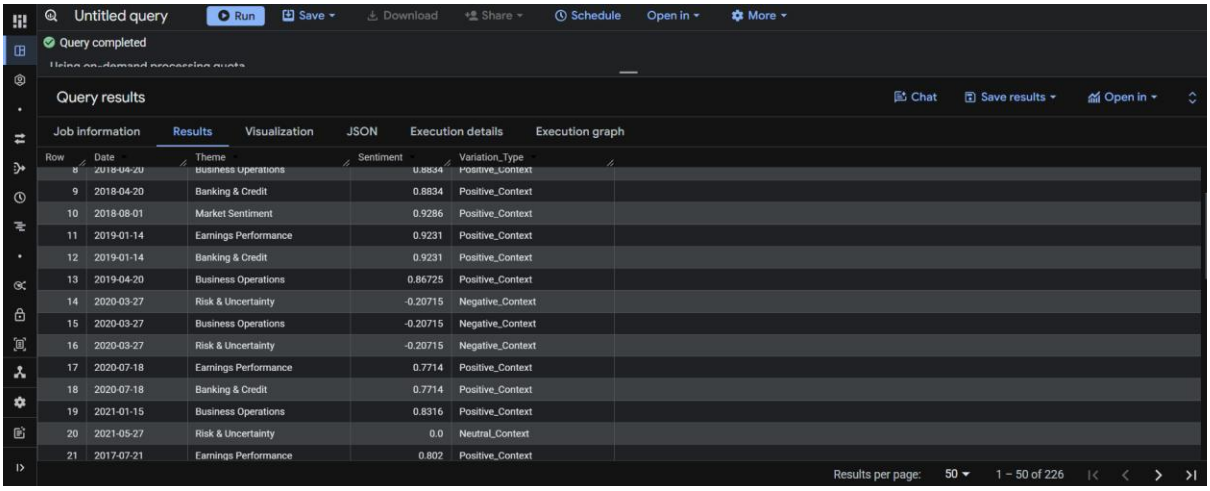
Overall, this question focuses on testing whether combining multiple signals can improve the strength of analysis, or whether adding weaker signals introduces noise that reduces the effectiveness of the stronger theme-based signal.

Sub Question 1 -

So, before combining all three at once, let me go slowly and understand. If I combine the theme with the sentiment, does the theme's signal change due to the sentiment, or not? In simple words, if a theme is generally seen as positive, does the sentiment change that overall signal if it points in a different direction? Will that also affect the overall signal or not?

So, there is this concept called context, which is important to understand. Entity phrases or themes are just key words in a sentence, but the conditions under which they are said are what we call context. For example, something like Capex or Growth Expansion is generally seen as positive in simple terms, but the context can give it a different meaning. For instance, growth expansion with too much debt can turn this phrase into a negative context.

So, this will be done by taking the themes view and combining it with the average sentiment view, joining them using date, and then assigning labels based on whether the context becomes positive or negative. This step focuses on understanding how sentiment adds context to themes, and whether combining them changes the direction or meaning of the original theme-based signal.



Row	Date	Theme	Sentiment	Variation_Type
8	2018-04-20	Business Operations	0.8834	Positive_Context
9	2018-04-20	Banking & Credit	0.8834	Positive_Context
10	2018-08-01	Market Sentiment	0.9286	Positive_Context
11	2019-01-14	Earnings Performance	0.9231	Positive_Context
12	2019-01-14	Banking & Credit	0.9231	Positive_Context
13	2019-04-20	Business Operations	0.86725	Positive_Context
14	2020-03-27	Risk & Uncertainty	-0.20715	Negative_Context
15	2020-03-27	Business Operations	-0.20715	Negative_Context
16	2020-03-27	Risk & Uncertainty	-0.20715	Negative_Context
17	2020-07-18	Earnings Performance	0.7714	Positive_Context
18	2020-07-18	Banking & Credit	0.7714	Positive_Context
19	2021-01-15	Business Operations	0.8316	Positive_Context
20	2021-05-27	Risk & Uncertainty	0.0	Neutral_Context
21	2017-07-21	Earnings Performance	0.802	Positive_Context

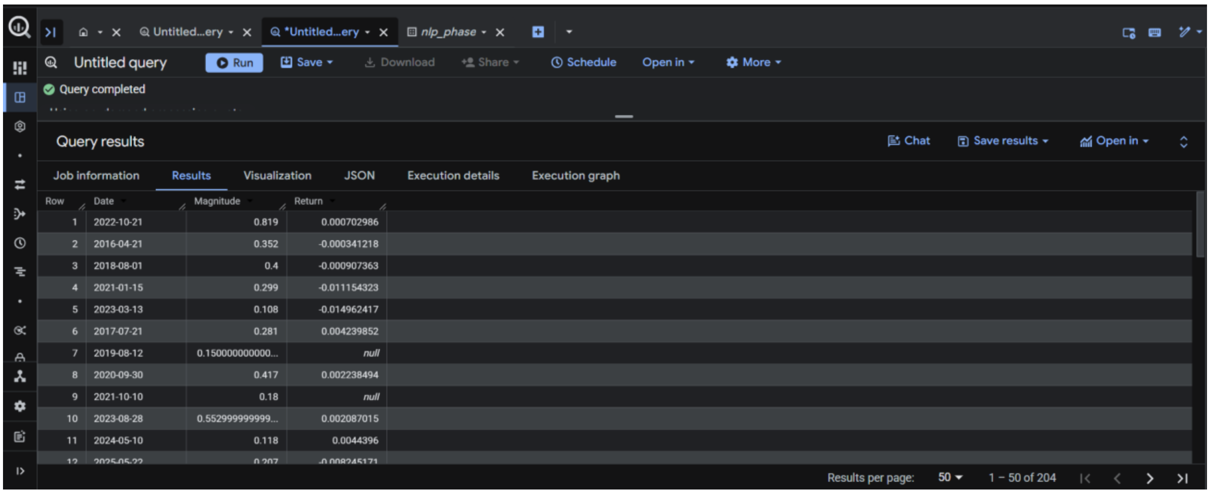
Sub Question 2 -

Now, one question that comes up is how the intensity of sentiment impacts market behaviour, and the reason is normally the strength of that sentiment. In formal terms, the magnitude matters. If the sentiment has strong intensity in any direction, it may be associated with stronger market movement.

So, in this, I am going to check the strength of the magnitude of the sentiments against the market return on that particular day. Previously, I used a common approach that grouped mild positive and strong positive reactions into a single bracket as positive sentiment, which may be one reason there was only around 50% accuracy in the sentiment analysis.

So this will be done by taking the magnitude value from the sentiment analysis and then taking the Nifty returns value and performing a LEFT JOIN on them. This will give only the news-date-based market return values. When I have done that, there are many null values, as news data can also come over weekends, where market returns are not available. This is expected, and such rows can be ignored during analysis.

The next step is to observe whether higher magnitude values are associated with larger market movement. So, this step focuses on understanding whether the strength of sentiment, rather than just its direction, has any meaningful relationship with the intensity of market movement.



Sub Question 3 -

Now, the main thing is to combine the sentiment, magnitude, and the entity with the themes, so that all the signals are combined, and to try to create a unified signal that can later be compared with market behaviour.

The concept here is a multi-signal system, which normally combines multiple signals to create more reliable information. The same idea is applied here by combining sentiment, entity, and theme signals to create a more reliable representation of the underlying information. This approach is better, as seen in the context concept, because each individual signal captures only partial information, and combining them provides a more complete view, though still not the full meaning of the sentence.

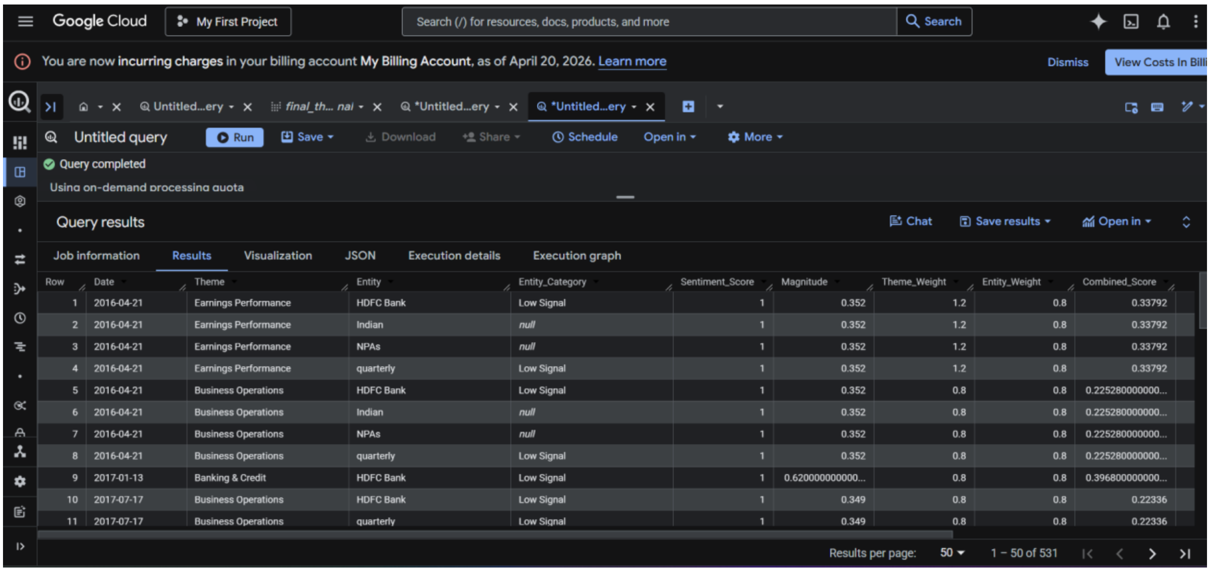
So how do I combine all these signals together? The main approach is to create a combined score by multiplying the individual scores of these signals, which I have done. Multiplying all of them gives a unified single score that can be used for further analysis.

For sentiment, I have assigned +1, -1, and 0 for positive, negative, and neutral tones, respectively. Even though they have their own direction, I normalized them because I am using the magnitude value to capture the strength.

Now, coming to the theme and entity scores, themes have been categorized as high, medium, and low based on their signal strength, and entities are categorized as high and low based on their signal category. The values assigned to them are 1.2 for high, 1 for medium, and 0.8 for low.

After implementation, there are some issues in the result table. Some entities do not have high or low signals because they were filtered out earlier, so these cases will need to be handled later. At this stage, the focus is on building the combined signal correctly, which can then be evaluated against market behaviour in the next step.

Overall, this step focuses on integrating multiple signals into a single combined score, aiming to capture a more complete representation of information, while also highlighting the practical challenges that arise when merging different signal layers.



Row	Date	Theme	Entity	Entity_Category	Sentiment_Score	Magnitude	Theme_Weight	Entity_Weight	Combined_Score
1	2016-04-21	Earnings Performance	HDFC Bank	Low Signal	1	0.352	1.2	0.8	0.33792
2	2016-04-21	Earnings Performance	Indian	null	1	0.352	1.2	0.8	0.33792
3	2016-04-21	Earnings Performance	NPAs	null	1	0.352	1.2	0.8	0.33792
4	2016-04-21	Earnings Performance	quarterly	Low Signal	1	0.352	1.2	0.8	0.33792
5	2016-04-21	Business Operations	HDFC Bank	Low Signal	1	0.352	0.8	0.8	0.225280000000...
6	2016-04-21	Business Operations	Indian	null	1	0.352	0.8	0.8	0.225280000000...
7	2016-04-21	Business Operations	NPAs	null	1	0.352	0.8	0.8	0.225280000000...
8	2016-04-21	Business Operations	quarterly	Low Signal	1	0.352	0.8	0.8	0.225280000000...
9	2017-01-13	Banking & Credit	HDFC Bank	Low Signal	1	0.620000000000...	0.8	0.8	0.396800000000...
10	2017-07-17	Business Operations	HDFC Bank	Low Signal	1	0.349	0.8	0.8	0.22336
11	2017-07-17	Business Operations	quarterly	Low Signal	1	0.349	0.8	0.8	0.22336

Sub Question 4 -

The main question is whether the combined score aligns with the market return, whether the multi-signal approach is better at capturing the mood of market behaviour, or whether sentiment and entity act as noise that diminishes the effect of the theme signal.

The main point to acknowledge is that, in NLP, much of the process is inherently subjective. The news data itself is subjective, and scarcity becomes a major issue when working over a long period like this 10-year dataset. Then there is the way the data is collected—what companies, what indicators, and what type of news are included. On top of that, how numerical values are assigned to text or unstructured data is also highly subjective.

This is different from ML, where models like regression, trees, ensemble methods, and classification are not inherently subjective in the same way; the results remain consistent, and only interpretation varies. But in NLP, subjectivity is more deeply embedded in the process itself, which can significantly influence the results. Even the way I combine all the signals together and create a combined score is a subjective decision, on which I am going to analyze the market behaviour.

So, I am going to compare the combined score with the Nifty return to check whether they move in the same direction. If they do, it is a match; otherwise, it is a mismatch. Then I calculate the percentage of matches to assess whether the approach is effective. To make the comparison clearer, I normalize both the combined score and the market return by assigning +1, -1, and 0, where positive values are +1, negative values are -1, and neutral values are 0.

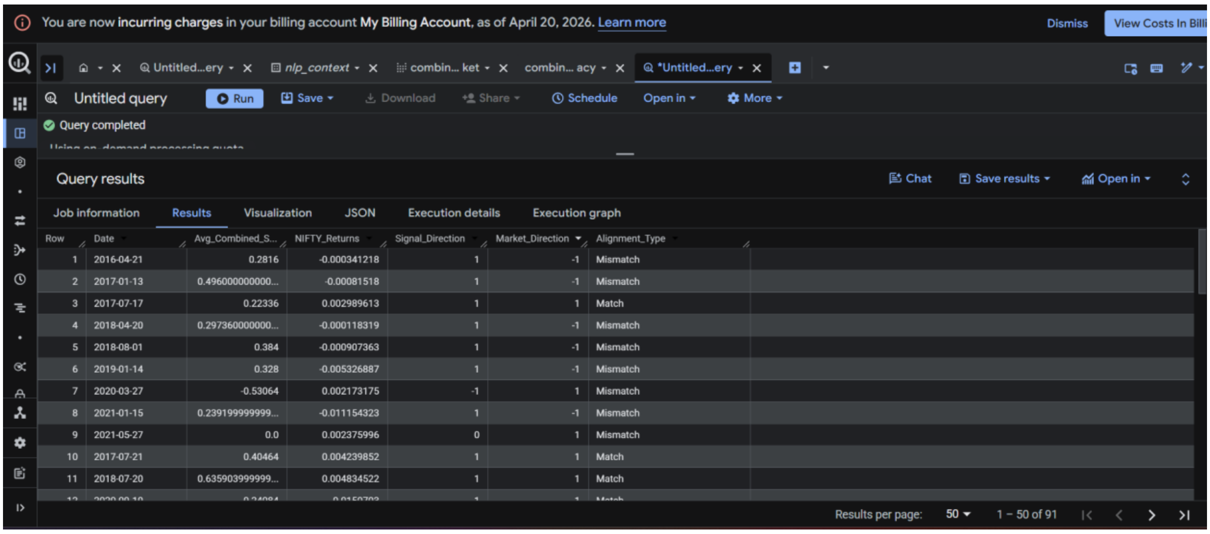
The implementation was quite challenging. First, I had to merge the news data into a single-date format, which is important because market returns are based on a single-date structure. Also, there can be a lot of noise when the same theme has multiple entities associated with it. To manage this, I took an average of the combined scores for signals on the same date.

However, the result was not strong. It showed that around 56% of the cases had mismatched directions, which is quite poor for a combined signal. The methodology itself, being subjective, can also be questioned, as I am using manual logic to combine the signals, which may be too simple for the complexity involved. I am also focused on using BigQuery SQL, which does not directly support advanced NLP functions, and I have avoided using more advanced tools like Google Vertex to keep the analysis within my understanding.

Then I tried to categorize the combined score into three levels to see if stronger signals perform better. I classified scores greater than 0.8 as high, between 0.8 and 0.3 as medium, and below that as low. The idea was that lower scores might represent more noise. The results showed that across all three categories, the mismatch rate remained high. Although the mismatch rate slightly decreased from low to high categories, the improvement was not significant enough to be considered useful.

Another issue was the limited distribution of strong signals, as there were only 4 rows in the high category. This makes the results less reliable and difficult to generalize. Also, the methodology itself can be questioned, but the pattern still suggests that with more data, the results could have been better understood.

In the end, combining these three signals did not yield a better result, and the outcome was worse than even the sentiment-based analysis. This happened because a lot of noise was introduced while combining these signals, and the subjective methodology may have further weakened the result. The main issue I see is that combining the signals, especially using simple averaging across multiple themes and entities, introduces noise and reduces the strength of the overall signal. So, the multi-signal approach in this form does not improve the understanding of market behaviour, as combining weaker signals with a stronger one introduces noise and reduces the overall effectiveness of the analysis.



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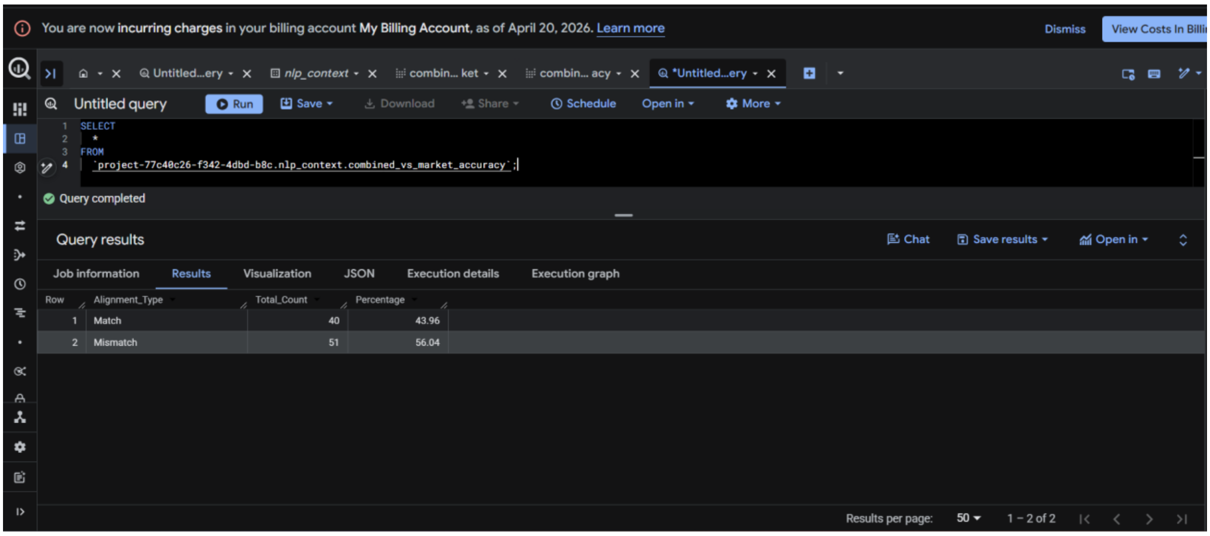
Open in
Job information
Results
Visualization
JSON
Execution details
Execution graph
Row | Date | Avg_Combined_S... | NIFTY_Returns | Signal_Direction | Market_Direction | Alignment_Type
1 | 2016-04-21 | 0.2816 | -0.000341218 | 1 | -1 | Mismatch
2 | 2017-01-13 | 0.496000000000... | -0.00081518 | 1 | -1 | Mismatch
3 | 2017-07-17 | 0.22336 | 0.002989613 | 1 | 1 | Match
4 | 2018-04-20 | 0.297360000000... | -0.000118319 | 1 | -1 | Mismatch
5 | 2018-08-01 | 0.384 | -0.000907363 | 1 | -1 | Mismatch
6 | 2019-01-14 | 0.328 | -0.005326887 | 1 | -1 | Mismatch
7 | 2020-03-27 | -0.53064 | 0.002173175 | -1 | 1 | Mismatch
8 | 2021-01-15 | 0.239199999999... | -0.011154323 | 1 | -1 | Mismatch
9 | 2021-05-27 | 0.0 | 0.002375996 | 0 | 1 | Mismatch
10 | 2017-07-21 | 0.40464 | 0.004239852 | 1 | 1 | Match
11 | 2018-07-20 | 0.635903999999... | 0.004834522 | 1 | 1 | Match
Results per page: 50
1 – 50 of 91



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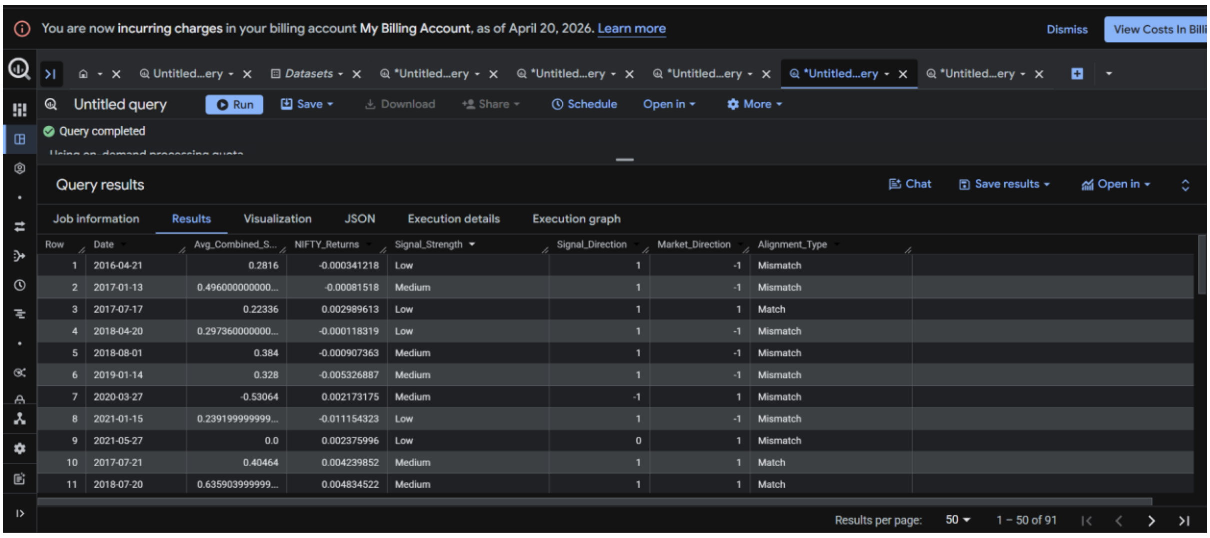
FROM
`project-77c40c26-f342-4dbd-b8c.nlp_context.combined_vs_market_accuracy`;
Query completed
Query results
Chat
Save results
Open in
Job information
Results
Visualization
JSON
Execution details
Execution graph
Row | Alignment_Type | Total_Count | Percentage
1 | Match | 40 | 43.96
2 | Mismatch | 51 | 56.04
Results per page: 50
1 – 2 of 2



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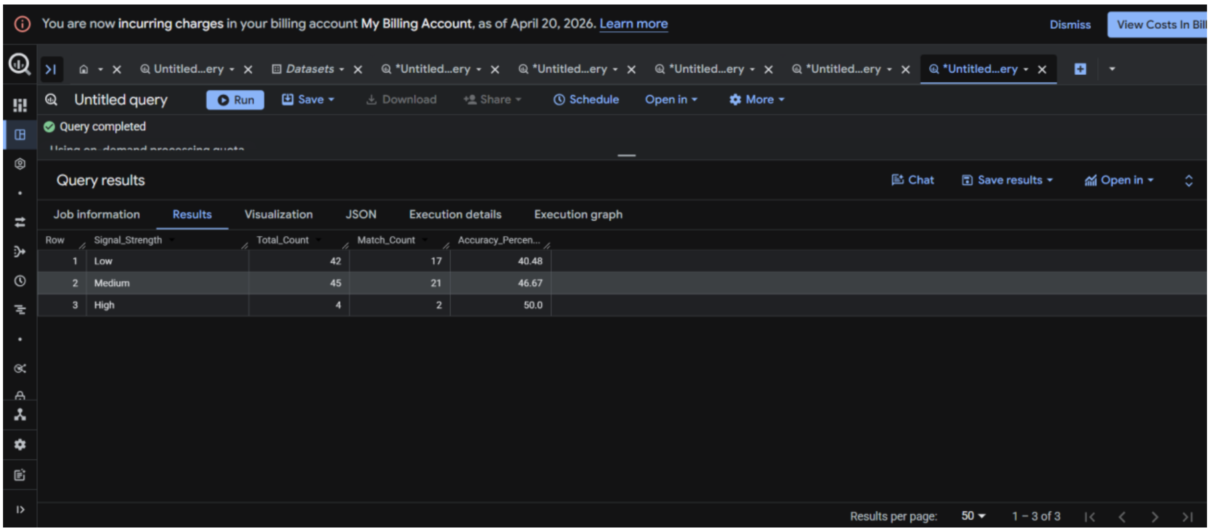
More
Query completed
Query results
Chat
Save results
Open in
Job information
Results
Visualization
JSON
Execution details
Execution graph
Row | Date | Avg_Combined_S... | NIFTY_Returns | Signal_Strength | Signal_Direction | Market_Direction | Alignment_Type
1 | 2016-04-21 | 0.2816 | -0.000341218 | Low | 1 | -1 | Mismatch
2 | 2017-01-13 | 0.496000000000... | -0.00081518 | Medium | 1 | -1 | Mismatch
3 | 2017-07-17 | 0.22336 | 0.002989613 | Low | 1 | 1 | Match
4 | 2018-04-20 | 0.297360000000... | -0.000118319 | Low | 1 | -1 | Mismatch
5 | 2018-08-01 | 0.384 | -0.000907363 | Medium | 1 | -1 | Mismatch
6 | 2019-01-14 | 0.328 | -0.005326887 | Medium | 1 | -1 | Mismatch
7 | 2020-03-27 | -0.53064 | 0.002173175 | Medium | -1 | 1 | Mismatch
8 | 2021-01-15 | 0.239199999999... | -0.011154323 | Low | 1 | -1 | Mismatch
9 | 2021-05-27 | 0.0 | 0.002375996 | Low | 0 | 1 | Mismatch
10 | 2017-07-21 | 0.40464 | 0.004239852 | Medium | 1 | 1 | Match
11 | 2018-07-20 | 0.635903999999... | 0.004834522 | Medium | 1 | 1 | Match
Results per page: 50
1 – 50 of 91



Answer of the fourth major question -

The answer is not what I was expecting. Combining all three signals did not yield the result I expected. The combined result is worse than what the sentiment signal alone provides. It also shows compatibility issues among the three signals when combined, as the negative effects outweigh the positive ones. This could also be due to how I am combining these three signals, but doing this poses a challenge in understanding how to view the dataset through the lens of NLP.

One important observation is that, while combining signals should ideally improve the overall understanding by capturing different aspects of the text, in this case it seems to be introducing more noise instead of clarity. Each signal — sentiment, theme, and entity — carries its own meaning, but when they are merged together without a strong structure or proper weighting logic, their individual strengths get diluted. This leads to a situation where weak or irrelevant signals interfere with stronger ones, reducing the overall effectiveness of the combined score.

Another point is that the averaging process used to combine multiple signals on the same date may be oversimplifying the problem. The market does not react in a linear or uniform way to all types of news, and by averaging, important variations and stronger signals may be getting cancelled out. This makes the final combined score less representative of the actual market sentiment or direction.

Additionally, the distribution of the signals also plays a role. There are very few strong signals compared to medium and low ones, which makes it difficult to clearly observe any strong pattern or improvement. Even though there is a slight improvement when moving from low to high signal strength, it is not significant enough to conclude that the combined signal is effective. This indicates that simply increasing the number of signals or combining them does not necessarily lead to better results.

Another important understanding is that NLP signals are not independent of each other. Sentiment, entity, and theme are all derived from the same underlying text, and combining them without considering their relationships can lead to overlapping information being counted multiple times. This further increases noise rather than improving clarity.

Overall, this analysis shows that combining multiple NLP signals is not straightforward and requires a more refined approach. The current method, which relies on simple aggregation and equal weighting, is not sufficient to capture the complexity of market behaviour. It highlights that the design of the signal and the method of combining them are critical factors, and without proper structure, combining signals can lead to worse outcomes than using a single well-defined signal. So, the result shows that adding more signals does not automatically improve understanding, and in this case, combining weaker signals with a stronger one introduces noise, reducing the effectiveness of the analysis instead of improving it.

Fifth Major Question of the Project -

After combining the three signals, the result did not produce any meaningful or useful signal for market returns. Even the sentiment signal, which showed around 50% alignment, performed better on its own than the combined signals (sentiment, entity, and themes). Among the three, the theme signal on its own provides a relatively stronger signal for understanding market behaviour compared to the other two.

This raises an important point that simply combining multiple signals does not necessarily improve the quality of the analysis, and in some cases, it can even weaken the overall signal due to the introduction of noise and overlapping information.

Now I am trying to see whether the company’s earnings data, which are released and announced by the company, have any meaningful impact on market behaviour. Earnings data represent the company’s own communication about its performance and future expectations, which can influence how market participants perceive the stock.

I am also going to check whether combining the theme signal with earnings data can provide a multi-signal that performs better than the individual effects of either. Since themes represent broader market narratives and earnings represent company-specific perspectives, combining these two may provide a more balanced signal compared to the earlier combination of sentiment, entity, and theme.

Overall, this question focuses on testing whether combining a stronger external signal (themes) with a company-driven signal (earnings) can create a more meaningful and reliable understanding of market behaviour compared to previous multi-signal attempts.

Sub Question 1 -

Now, the main focus is on the earnings text data of the companies, which will be used for the analysis in this NLP project. Before doing the analysis, I first need to extract this unstructured text data into meaningful scores that can be used for analysis. As I have already done at various levels in this project, I am going to use NLP functions for this, mainly sentiment functions for now, to extract the sentiment score (or direction score) and the magnitude (or strength score).

Extracting these numerical values from the earnings text data will be done using Python, as BigQuery SQL does not directly support NLP functions.

Earnings data represents the information provided by the company and its management about current operations and performance, as well as future outlook. These are among the main drivers of market behaviour and how the company is perceived in the near term.

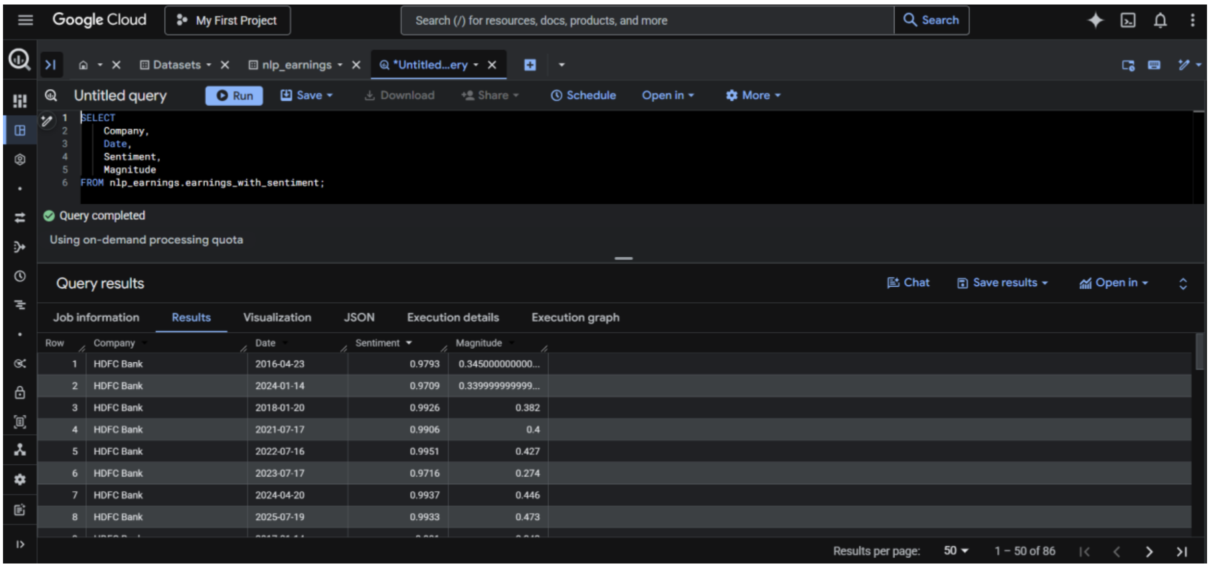
I have already explained the sentiment functions used in earlier analysis. The sentiment score (or direction score) indicates the direction of the text, ranging from -1 to +1. Values closer to -1 indicate a negative direction, closer to +1 indicate a positive direction, and values around 0 represent neutral sentiment. The magnitude (or strength) score shows the intensity of the text.

I am going to use the same approach here, using the Vader sentiment analyzer in Python, which I have used before as well. The logic remains the same. Vader is a rule-based NLP model that uses predefined rules and a dictionary of words to assign values when data is processed. It breaks down each sentence into words and provides four scores: positive, negative, neutral, and compound.

The compound score reflects the overall emotion in the sentence, which I use as the sentiment score for earnings analysis, as it gives a complete picture of the overall sentiment and direction. For magnitude, I combine the positive and negative scores, which helps to understand the intensity of the words in the sentence.

After extracting these values, I created a view from the dataset for further analysis. This process converts unstructured earnings text into structured numerical features (sentiment and magnitude), which are stored as a view in BigQuery and serve as the base dataset for further analysis.

Overall, this step focuses on converting earnings-related text into measurable sentiment signals, creating a structured foundation to analyze whether company-driven narratives have any meaningful impact on market behaviour.



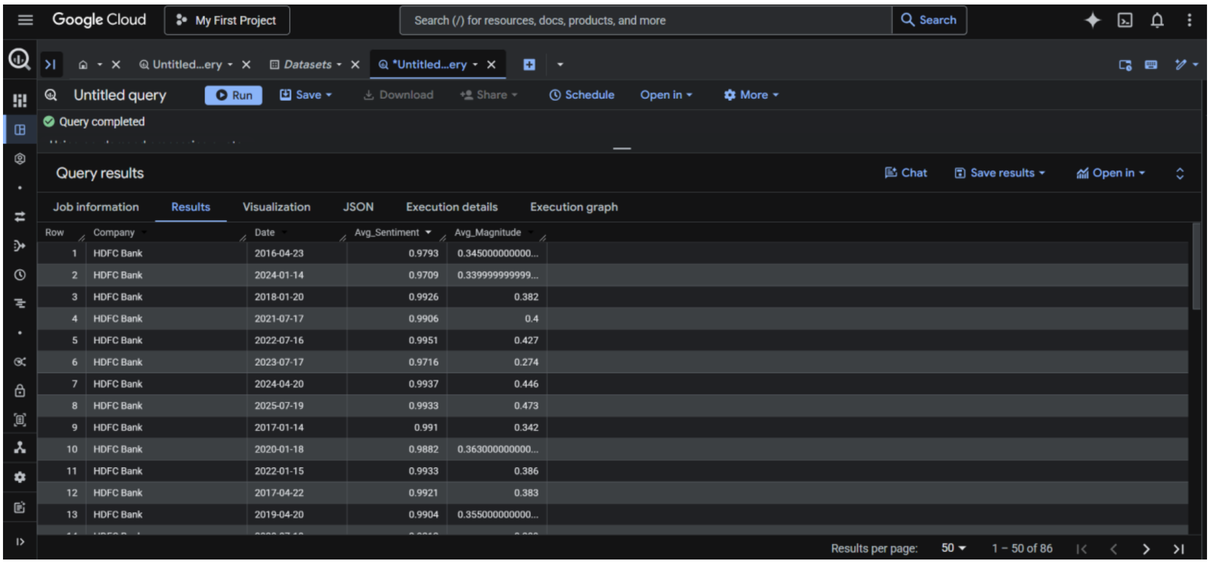
Sub Question 2 -

Now that the sentiments from the earnings data have been extracted, it's time to create a uniform company-level signal. This means creating sentiment scores with respect to date and company, which will show how the sentiment of that company behaves at each earnings event. Since earnings data comes at specific intervals, this can later be used to compare how that stock performs around those events. It also provides another application to compare the company's earnings sentiment with its market return and understand how the market reacts to it. But for now, this step helps in understanding the direction of company-level sentiment at each earnings event.

This aggregation at the company level will be done using average sentiment and average magnitude, as it makes it easier to understand the overall impact and provides a more realistic view of the earnings data.

The view contains the columns Company, Date, Avg_Sentiment, and Avg_Magnitude as the schema, which will be useful for further analysis. These are single-date fields that can be directly joined with market returns in later steps. This creates one sentiment signal per company per earnings date.

In the end, this step standardizes earnings sentiment at the company-date level, creating a clean and comparable signal that can be directly linked with market performance in further analysis.



Sub Question 3 -

Now that the earnings data has been aggregated, the main analysis will focus on the impact of these earnings data on market movement or market behaviour. As this is again a signal and outcome perspective, the earnings data will work as a factor that can influence the outcome, which is the market return. Analysing that will be the main work for now.

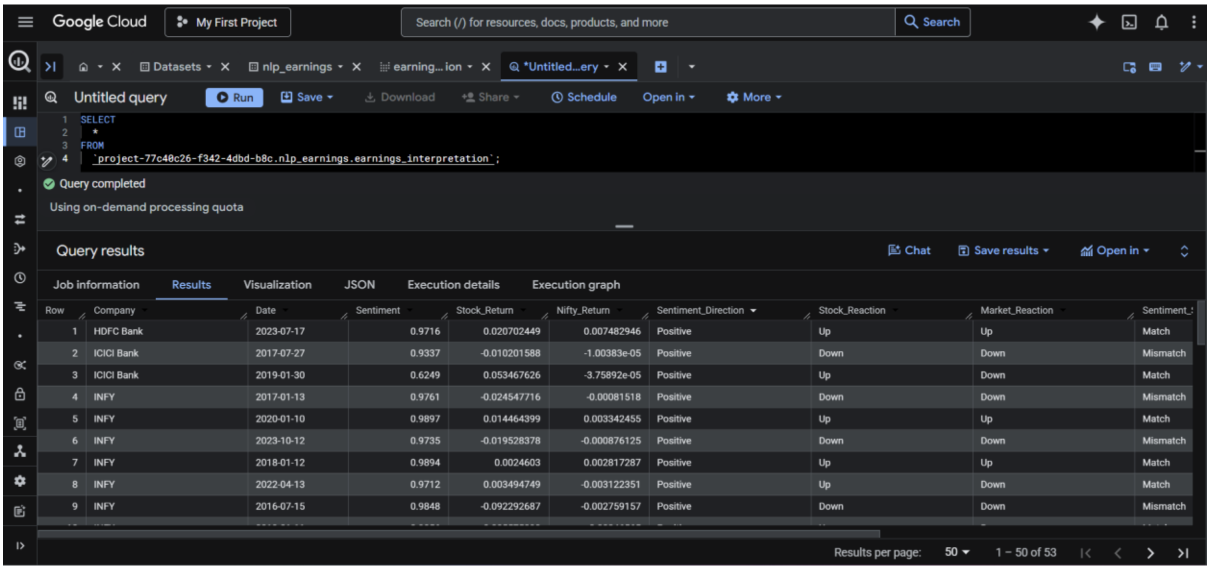
The outcome remains focused based on the analysis and understanding of the earnings data. The main point is to determine whether these earnings data have any impact on their respective stocks, so the outcome in this step is the stock-level movement only.

The implementation starts with extracting these stock returns into a separate view, which will then be combined with the earnings sentiment view using a LEFT JOIN. It is also important to note that some earnings data come on weekends or days when market returns are not available, so those instances are excluded from the analysis.

In the main analysis, I observe that out of 16 HDFC earnings news items, only one was released when the market was open, while the others were released when the market was closed. This makes it very difficult to assess the actual impact. With 10 years of data, extracting only 16 earnings news items is quite limited, which highlights the issue of data scarcity over a longer period.

The same issue was observed for ICICI, but for the other three companies, their earnings news appeared within market news, making it slightly easier to analyse them. Infosys (INFY) shows the best match percentage among the companies, which indicates that its stock movement aligns better with earnings sentiment. In contrast, Reliance and TCS show a more mixed or half-and-half pattern, which makes it difficult to draw clear conclusions.

In the end, the next step is to combine this analysis with another strong signal, which is themes, and understand whether these two signals complement each other to form a more stable and meaningful signal, or whether the same limitations appear as seen earlier when combining sentiment, entity, and theme signals together. So, this step shows that earnings sentiment has some alignment with stock-level movement in specific cases, but the impact is limited due to data scarcity and timing issues, making it difficult to draw strong and consistent conclusions.



```sql
SELECT
  *
FROM
  `project-77c40c26-f342-4dbd-b8c.nlp_earnings.earnings_interpretation`;
```

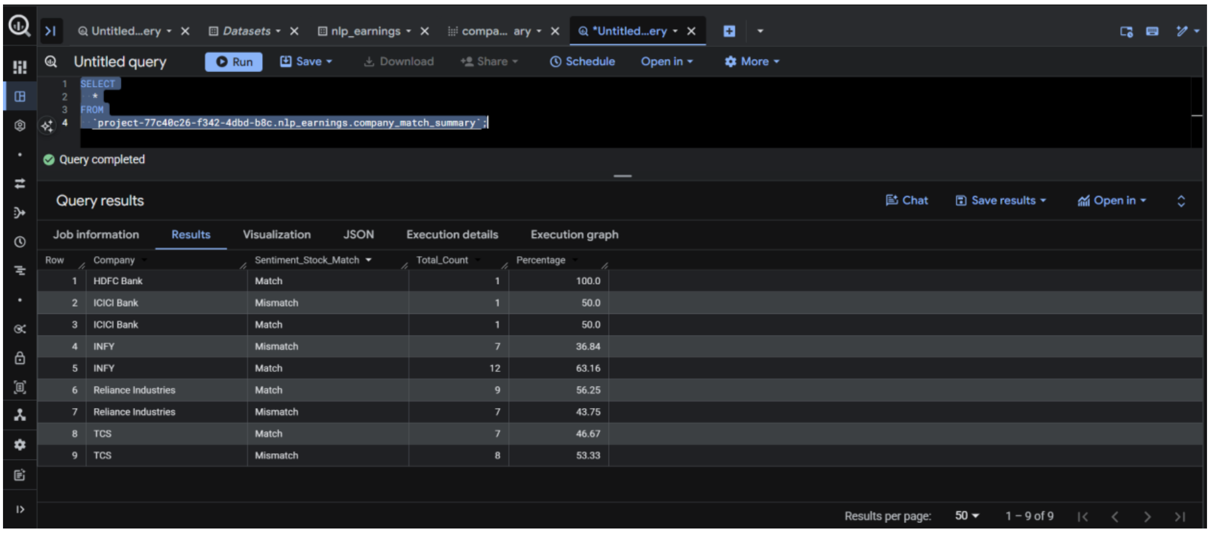
Query completed

Using on-demand processing quota

Query results

Row	Company	Date	Sentiment	Stock_Return	Nifty_Return	Sentiment_Direction	Stock_Reaction	Market_Reaction	Sentiment_:
1	HDFC Bank	2023-07-17	0.9716	0.020702449	0.007482946	Positive	Up	Up	Match
2	ICICI Bank	2017-07-27	0.9337	-0.010201588	-1.00383e-05	Positive	Down	Down	Mismatch
3	ICICI Bank	2019-01-30	0.6249	0.053467626	-3.75892e-05	Positive	Up	Down	Match
4	INFY	2017-01-13	0.9761	-0.024547716	-0.00081518	Positive	Down	Down	Mismatch
5	INFY	2020-01-10	0.9897	0.014464399	0.003342455	Positive	Up	Up	Match
6	INFY	2023-10-12	0.9735	-0.019528378	-0.000876125	Positive	Down	Down	Mismatch
7	INFY	2018-01-12	0.9894	0.0024603	0.002817287	Positive	Up	Up	Match
8	INFY	2022-04-13	0.9712	0.003494749	-0.003122351	Positive	Up	Down	Match
9	INFY	2016-07-15	0.9848	-0.092292687	-0.002759157	Positive	Down	Down	Mismatch

Results per page: 50 1 – 50 of 53



```sql
SELECT
  *
FROM
  `project-77c40c26-f342-4dbd-b8c.nlp_earnings.company_match_summary`;
```

Query completed

Query results

Row	Company	Sentiment_Stock_Match	Total_Count	Percentage
1	HDFC Bank	Match	1	100.0
2	ICICI Bank	Mismatch	1	50.0
3	ICICI Bank	Match	1	50.0
4	INFY	Mismatch	7	36.84
5	INFY	Match	12	63.16
6	Reliance Industries	Match	9	56.25
7	Reliance Industries	Mismatch	7	43.75
8	TCS	Match	7	46.67
9	TCS	Mismatch	8	53.33

Results per page: 50 1 – 9 of 9

Sub Question 4 -

This will probably be the last major question in this project. I have already tried multi-signal analysis before by combining entity, sentiment, and themes together, and the result was not good. So now I am combining only two signals. The first is the strongest signal I have, which is the theme signal, and the second is the earnings sentiment signal from this analysis. The main question is whether combining these two dilutes the theme signal or makes it stronger.

One of the major challenges I have faced from the start of this project is the scarcity of data, which has created a lot of limitations. The methodology I used earlier to understand unstructured text also did not work well. Combining signals using simple averaging or direct multiplication resulted in poor outcomes. This shows that unstructured text analysis is much more complex than ML-based data, where structured models and tools are readily available.

BigQuery SQL directly supports ML models, but it does not support NLP functions. So, these functions have to be implemented either using tools like Google Vertex, which are more complex, or through Python, where manual logic and understanding need to be built from scratch. This project has therefore been more about learning these limitations and understanding the nature of NLP rather than just achieving strong results. In my previous ML project, implementation was easier but analysis was challenging, whereas in NLP, both extraction and analysis are challenging due to the lack of direct support.

The main thing to understand about these two signals is how they operate. The earnings signal reflects how the company presents itself to the public, which is mostly from the company's own perspective. In contrast, the theme signal reflects the broader perception of the market, based on economic conditions, global events, and overall sentiment in the system. If the company's communication aligns with the broader market perception, both signals will move in the same direction.

One important point to note is that most earnings data is biased toward the positive side, as companies tend to present their performance in a favorable way. Because of this, mismatches can occur when the broader market perception, captured by themes, does not align with the company's communication.

I combined the theme signal and the earnings signal using defined categories. If the earnings sentiment score is greater than 0.5, it is considered strong, and for the theme signal, if the average return is greater than 0, it is considered strong; otherwise, it is weak. Based on this, I created four categories: strong positive, where both signals are strong; conflicting positive, where earnings are strong but themes are weak; conflicting negative, where earnings are weak but themes are strong; and strong negative, where both are weak.

One thing to understand here is that earnings data inherently carries a positive bias, as companies tend to control how information is communicated. This makes it difficult to interpret neutrality or negativity in earnings sentiment. In my dataset, most of the earnings sentiment scores are positive, which reflects this limitation.

The analysis shows that around 82.5% of the observations fall under strong positive. However, this needs to be interpreted carefully, as it is influenced by the positive bias in earnings data. The more important observation is that around 15% fall under conflicting positive, which highlights the gap between company communication and broader market conditions. At the same time, the high strong-positive percentage also suggests that in many cases, themes and earnings are aligned.

Data scarcity also plays a role here, as only around 40 earnings-related data points were available over the 10-year period. This limits the strength of the conclusions, although the results still provide some level of insight.

Now, when I tested this combined signal against market behaviour, I performed two analyses: one with stock-level movement and another with index-level movement. The results show better alignment with stock behaviour, with around 69% alignment. In my opinion, this is a reasonable outcome, given the bias in earnings data and the limitations in capturing all relevant information.

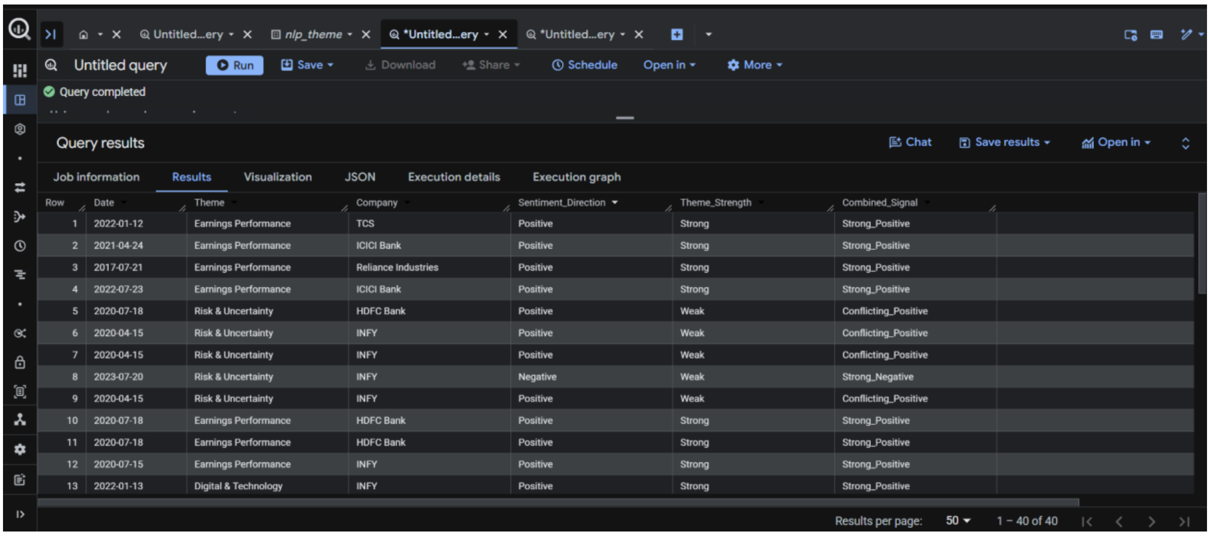
The scarcity of data remains the biggest limitation of this project. Unlike ML, where historical quantitative data is easily available, earnings data is qualitative and difficult to collect consistently. Still, achieving close to 70% alignment with stock returns suggests that the combined signal has some level of usefulness at the company level.

However, there are structural issues as well. For example, only Infosys and TCS earnings data were available on trading days, while for other companies, most earnings data fell on weekends or holidays, making it difficult to observe actual market reactions.

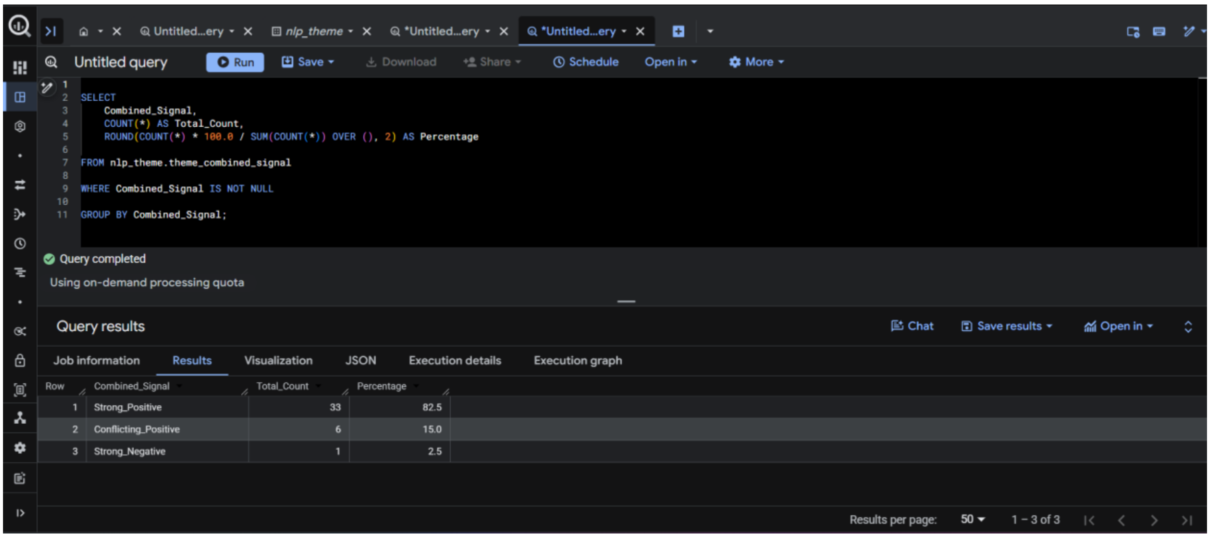
When I compare this with overall market behaviour using index returns, the results remain consistent with earlier findings in the project. The match percentage is just above 51%, which is close to a random outcome. This consistency shows that capturing overall market behaviour is much more complex and cannot be explained easily using these signals.

This also highlights an important learning: understanding market behaviour is not straightforward, and using earnings and theme signals without sufficient data depth is not enough to capture broader market movements.

Overall, the combined signal shows better alignment at the stock level but fails to explain broader market behaviour, which reinforces the idea that while structured signals can help at a micro level, understanding overall market movement requires deeper data, better methodology, and more robust signal design.

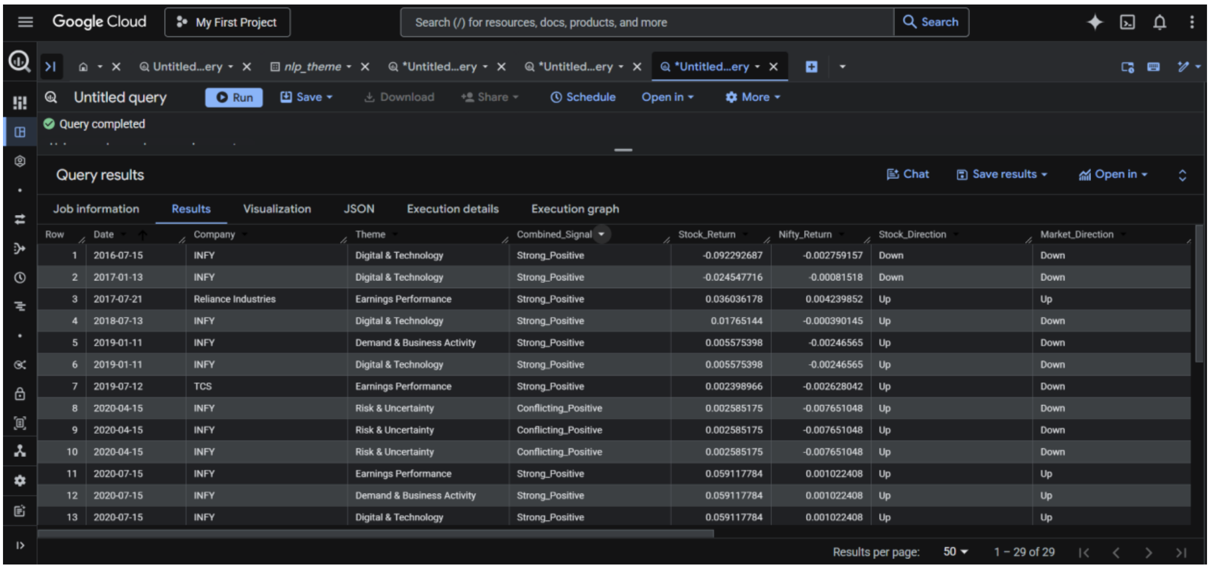


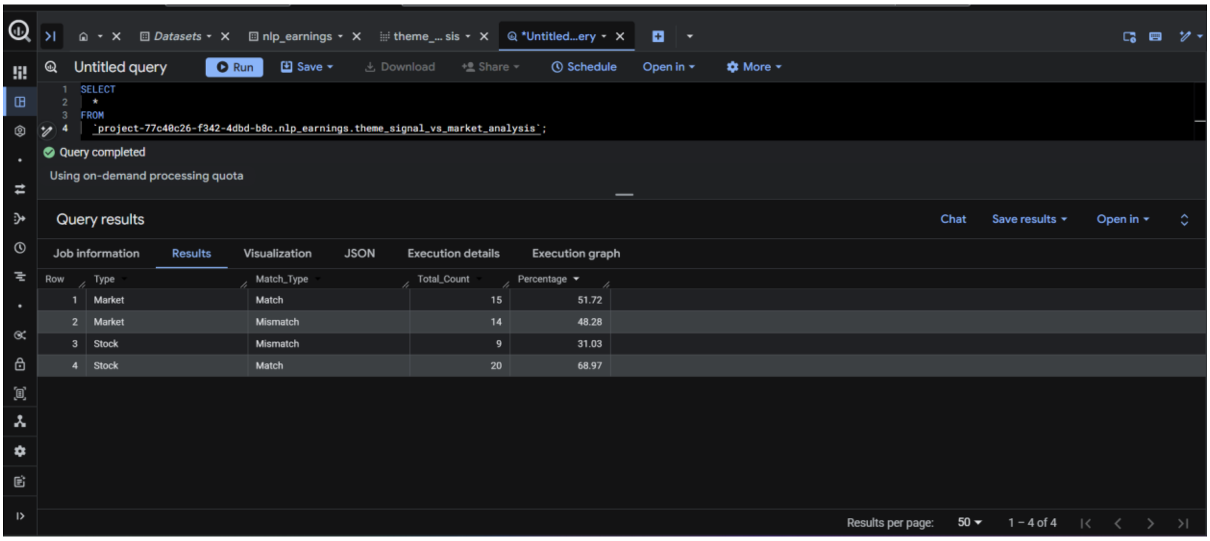
Row	Date	Theme	Company	Sentiment_Direction	Theme_Strength	Combined_Signal
1	2022-01-12	Earnings Performance	TCS	Positive	Strong	Strong_Positive
2	2021-04-24	Earnings Performance	ICICI Bank	Positive	Strong	Strong_Positive
3	2017-07-21	Earnings Performance	Reliance Industries	Positive	Strong	Strong_Positive
4	2022-07-23	Earnings Performance	ICICI Bank	Positive	Strong	Strong_Positive
5	2020-07-18	Risk & Uncertainty	HDFC Bank	Positive	Weak	Conflicting_Positive
6	2020-04-15	Risk & Uncertainty	INFY	Positive	Weak	Conflicting_Positive
7	2020-04-15	Risk & Uncertainty	INFY	Positive	Weak	Conflicting_Positive
8	2023-07-20	Risk & Uncertainty	INFY	Negative	Weak	Strong_Negative
9	2020-04-15	Risk & Uncertainty	INFY	Positive	Weak	Conflicting_Positive
10	2020-07-18	Earnings Performance	HDFC Bank	Positive	Strong	Strong_Positive
11	2020-07-18	Earnings Performance	HDFC Bank	Positive	Strong	Strong_Positive
12	2020-07-15	Earnings Performance	INFY	Positive	Strong	Strong_Positive
13	2022-01-13	Digital & Technology	INFY	Positive	Strong	Strong_Positive




```sql
SELECT
    Combined_Signal,
    COUNT(*) AS Total_Count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS Percentage

FROM nlp_theme.theme_combined_signal

WHERE Combined_Signal IS NOT NULL

GROUP BY Combined_Signal;
```

Row	Combined_Signal	Total_Count	Percentage
1	Strong_Positive	33	82.5
2	Conflicting_Positive	6	15.0
3	Strong_Negative	1	2.5





Answer of the fifth major question -

The main question here is whether the company's earnings data have any meaningful impact on market behaviour, and whether combining this earnings signal with the theme signal can create a stronger multi-signal compared to using them individually.

While working on this, the first thing I observed is that the earnings signal itself is not very neutral in nature. Most of the earnings communication is on the positive side, as companies generally try to present their performance in a favourable way. This creates a clear positive bias in the earnings sentiment, which becomes one of the biggest challenges in analysing it properly.

When I analysed the earnings signal on its own, the results were not very strong. The alignment with stock behaviour was limited, and with overall market behaviour, it again remained around a mid-level outcome. This shows that earnings data alone are not sufficient to explain market movements in a consistent way, especially given the limited data and biased sentiment structure.

After that, I combined the earnings signal with the theme signal, as the theme signal was already the strongest among all the signals I had. The idea behind this was that if the company's perspective and the broader market perception align, then the combined signal should become stronger and more meaningful.

From the combined signal, I observed that most of the cases fall under the strong positive category, which again reflects the positive bias in the earnings data. At the same time, the presence of conflicting cases shows that the company's communication and the broader market perception are not always aligned, which is an important insight in itself.

When I checked this combined signal against stock behaviour, the alignment was relatively better compared to the individual signals. This suggests that combining the theme and earnings signals provides some improvement at the company level, where stock movements are more directly influenced by company-specific information.

However, when I analysed the same combined signal with overall market behaviour, the results again remained around the 50% level. This shows that even after combining the two signals, the model is not able to capture broader market movements effectively, which has been a consistent observation throughout the project.

One of the main reasons behind this is the limitation of the data. The number of earnings data points is quite limited over the 10-year period, and many of them fall on non-trading days, which reduces the usable data further. Along with that, the subjective nature of earnings communication and the bias in sentiment also weaken the reliability of the signal.

Another important understanding is that market behaviour depends on many other factors beyond earnings and themes, such as macroeconomic conditions, global events, liquidity, and broader investor sentiment. Capturing all of this through limited textual data becomes very difficult.

Overall, combining the theme signal with the earnings signal does provide a slight improvement at the stock level, but it does not significantly improve the understanding of broader market behaviour. The results remain limited, and the signal does not become strong enough for reliable market-level interpretation. So, this analysis shows that while combining earnings and theme signals adds some value at the company level, it is still not sufficient to explain market behaviour comprehensively, highlighting that data quality, signal design, and broader context are critical for meaningful NLP-based financial analysis.

Answer to the Problem Statement

The main objective of this project was to understand whether adding unstructured data such as news, themes, and earnings information can provide meaningful additional insight into market behaviour, beyond what traditional structured market data can offer.

From the beginning, the idea was not to build a complex model, but to understand whether these unstructured signals can actually explain or align with market movement in a meaningful way.

As I progressed through the different stages of the project, starting from sentiment, then moving to entities, and then themes, one clear pattern emerged. Not all signals extracted from unstructured data are equally useful. Sentiment, which focuses only on the tone of the news, turned out to be a weak signal, as it showed almost a 50-50 alignment with market returns. This indicates that tone alone does not capture the real drivers of market behaviour.

When I moved to entity-based analysis, the expectation was that focusing on important subjects of the news might provide a better signal. However, the results again showed inconsistency. Even after filtering for important entities using salience and frequency, most entities did not show a stable or consistent relationship with market returns. This suggests that individual elements of the news, even if important, may still act as noise when analysed in isolation.

The most meaningful improvement came when I shifted the focus to themes. Unlike sentiment and entities, themes capture a broader narrative by combining multiple related signals into a single structured idea. The results from theme analysis showed relatively stronger and more interpretable patterns, both in positive and negative directions. This indicates that market behaviour is better explained at a broader narrative level rather than through isolated signals.

However, when I attempted to combine multiple signals together, the results did not improve. In fact, combining sentiment, entity, and theme signals reduced the overall effectiveness. This highlights an important point that simply combining multiple weak or partially useful signals does not necessarily create a stronger signal. Instead, it introduces more noise and dilutes the impact of the stronger component.

In the final stage, I analysed earnings data, which represents the company's own communication and perspective. One clear observation here is that earnings data is inherently biased towards positivity, as companies tend to present their performance in a favourable way. This bias affects the reliability of sentiment extracted from earnings text.

When earnings signals were analysed independently, they did not show strong alignment with market behaviour. However, when combined with theme signals, there was a noticeable improvement at the stock level, where alignment reached around 70%. This suggests that when company-level communication aligns with broader market themes, the signal becomes more meaningful at the individual stock level.

At the market level, however, the results remained around the 50% range throughout the project. This consistency indicates that broader market movement is influenced by many factors that are not fully captured by the limited unstructured data used in this project.

One of the most important learnings from this project is the role of data quality and availability. The scarcity of news and earnings data over a long time period significantly limits the strength and reliability of the signals. In contrast to structured market data, unstructured data is harder to collect, more subjective in nature, and more sensitive to how it is processed and interpreted.

Overall, the project shows that unstructured data can provide additional context and some level of insight into market behaviour, especially when analysed at a broader thematic level. However, it is not sufficient on its own to consistently explain or predict market movement, particularly at the index level.

The results suggest that while themes can act as a relatively stronger signal and earnings data can add value at the company level, the effectiveness of these signals depends heavily on data quality, depth, and the methodology used. Without sufficient data and a well-defined structure, combining multiple NLP signals can lead to more noise rather than clarity.

Overall, the project shows that unstructured data is useful for adding context and improving understanding at a conceptual level, but it is not a standalone solution for explaining market behaviour, and its effectiveness depends on careful design, sufficient data, and controlled methodology.

Summary of the Project

This project focuses on understanding whether unstructured data such as news, themes, and earnings can add meaningful insight into market behaviour. Instead of building a complex model, the approach is centred on analysing different signals extracted from text and evaluating their relationship with market movement. The analysis follows a structured process where each signal is tested individually and then in combination to understand its effectiveness, with the objective of identifying which signals provide useful understanding of market behaviour.

Key Findings

The main objective of this project was to understand whether unstructured data such as news, themes, and earnings can provide meaningful insight into market behaviour beyond traditional structured data.

From the overall analysis, one clear pattern emerged. Not all signals extracted from unstructured data are equally useful. Sentiment, which captures only the tone of the news, turned out to be a weak signal, showing around a 50% alignment with market movement. This indicates that tone alone is not sufficient to explain market behaviour.

Entity-based analysis improved the understanding slightly by focusing on the subjects of the news. However, most entities did not show a consistent or stable relationship with market returns, which suggests that individual elements of the news often act as noise when analysed in isolation.

The most meaningful results came from theme-based analysis. Themes capture broader narratives by combining multiple related signals into one structure. These showed clearer and more interpretable patterns in both positive and negative directions, indicating that market behaviour is better understood at a narrative level rather than through isolated signals.

When earnings data was analysed, it showed a strong positive bias, as companies tend to present their performance favourably. On its own, earnings data did not provide strong alignment with market behaviour. However, when combined with theme signals, there was a noticeable improvement at the stock level, with alignment reaching around 70%.

At the market level, however, all signals remained close to a 50% outcome, showing that broader market movement is not easily explained using these signals alone

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What Worked

The most effective part of the project was the shift from isolated signals to broader thematic understanding.

Theme-based analysis worked relatively well because it captures how multiple factors interact together, which is closer to how market behaviour actually evolves. It reduced the noise seen in sentiment and entity analysis and provided more interpretable insights.

Combining theme signals with earnings data also worked to some extent at the company level. When company communication aligned with broader market themes, the signal became more meaningful, and this was reflected in better alignment with stock movement.

Another important aspect that worked well was the structured approach of breaking the problem into smaller questions and analysing each signal independently before combining them. This helped in understanding where the actual strength and weakness of each signal lies.

What Did Not Work

Sentiment and entity signals did not perform well as standalone indicators. Sentiment lacked depth, and entity analysis, even after filtering, remained inconsistent and noisy.

The multi-signal approach, where sentiment, entity, and theme were combined together, did not improve the results. Instead, it introduced more noise and reduced the effectiveness of the stronger theme signal. This shows that combining multiple weak signals does not necessarily create a stronger one.

Earnings data also had limitations due to its positive bias and limited availability. Many earnings events occurred on non-trading days, reducing the usable data further.

At the market level, none of the signals — individual or combined — were able to move beyond a mid-level outcome. This indicates that market behaviour depends on many external factors that are not captured in the dataset.

Final Takeaway

The project shows that unstructured data can add meaningful context to market behaviour, but it is not sufficient to explain or predict market movement in a consistent way.

Themes emerge as the most useful signal because they capture broader narratives, while earnings data can add value at the company level when aligned with these themes.

However, the effectiveness of these signals depends heavily on data quality, availability, and how the signals are constructed. Limited data, bias in text, and subjective methodology reduce the strength of the analysis.

In conclusion, unstructured data should not be seen as a replacement for structured data, but as a complementary layer that provides context. It helps in understanding *why* the market behaves in a certain way, but not reliably *what* the market will do next.
